BITS PILANI | GOA CAMPUS

PROJECT ANTARA

Mission Concept Review

Narrative Edition

3U CubeSat for In-Situ Characterization of the
South Atlantic Anomaly Radiation Environment

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1 Mission Context, Objectives, and Success Criteria

1.1 Executive Summary

Project ANTARA is a 3U CubeSat mission at BITS Pilani, K.K. Birla Goa Campus, dedicated to the in-situ characterization of the South Atlantic Anomaly (SAA) radiation environment. Operating in a 600 km Sun-Synchronous Orbit (97.6° inclination), the spacecraft carries a custom Flux Particle Detector Unit (FPDU) payload to measure unidirectional differential proton flux across the 5–100 MeV range in a proposed multi-channel energy reconstruction.

The mission is structured around three tiers of success that scale spatial resolution, coverage, and scientific ambition. Beyond its scientific objective, ANTARA serves as a student satellite training platform, exposing undergraduate students to the complete spacecraft development lifecycle from concept through operations.

This document represents the Mission Concept Review (MCR) package for Project ANTARA, consolidating mission requirements, architecture, orbit design, operational concept, subsystem design status, and simulation results.

1.2 Mission Need

The ANTARA mission has been conceptualized to address a combination of scientific, educational, and national-level needs associated with the growing importance of space-based radiation studies and indigenous small-satellite development. Modern spacecraft operating in Low Earth Orbit (LEO) are increasingly exposed to harsh radiation environments that can significantly affect mission reliability, onboard electronics, detector systems, and long-term spacecraft survivability. Among these environments, the South Atlantic Anomaly (SAA) represents one of the most severe and scientifically significant radiation regions encountered by satellites in LEO.

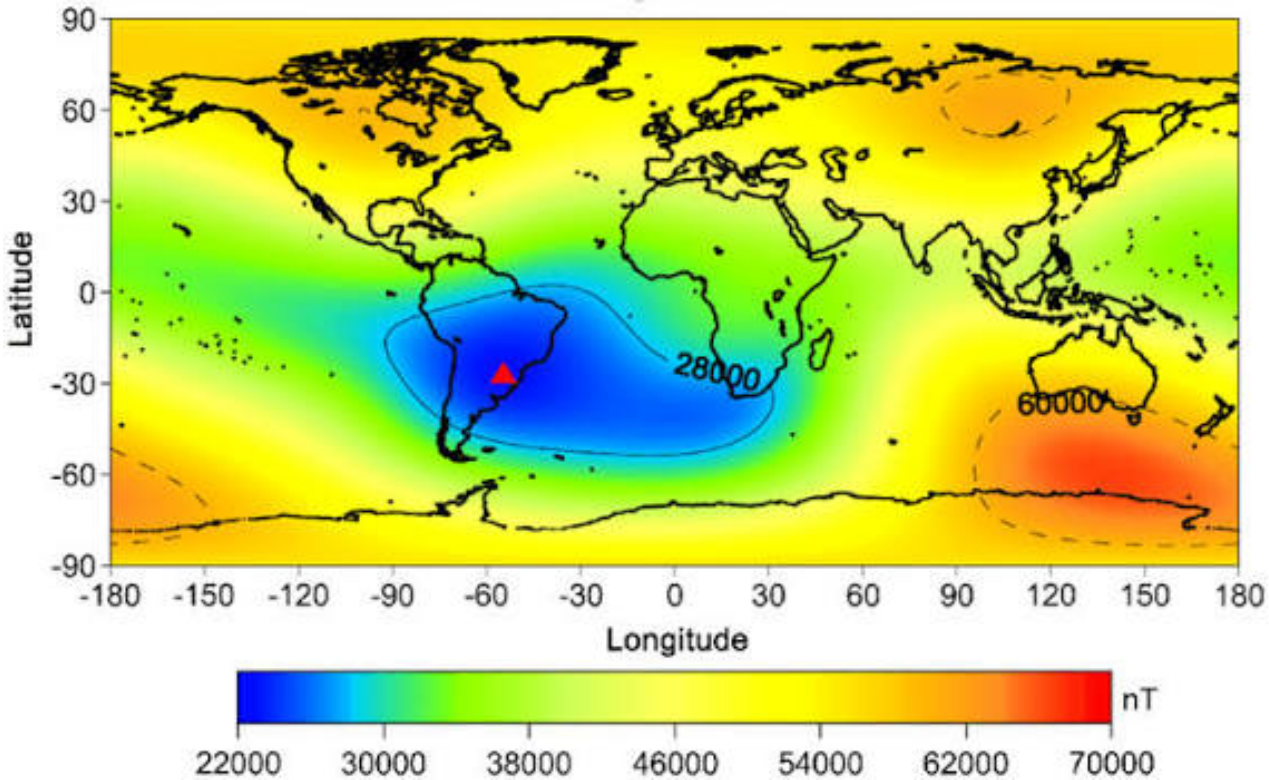


Figure 1: South Atlantic Magnetic Anomaly (year 2000; source image from original draft).

Although several trapped-radiation models such as AP8 and AP9 currently exist, substantial uncertainties remain regarding temporal proton-flux variability, storm-time behavior, directional anisotropies, and localized radiation structures within the SAA. Existing models are primarily statistical and often lack sufficient real-time data required for modern small-satellite missions and future radiation-aware spacecraft operations.

The ANTARA mission seeks to contribute toward closing these gaps through direct in-situ measurement of high-energy proton populations using a dedicated CubeSat operating in a Sun-Synchronous Orbit. The project also enables students to gain practical experience in CubeSat designing and modelling.

1.2.1 Educational Motive

One of the primary motivations behind ANTARA is the creation of an educational and research platform capable of providing hands-on experience in real-world spacecraft development.

The mission enables participating students to experience all major phases of aerospace project development. Unlike isolated academic laboratory experiments, CubeSat missions introduce students to highly constrained engineering environments where mass, power, thermal limits and others interact simultaneously. This systems-level complexity is one of the most important educational benefits of the ANTARA mission.

1.2.2 Scientific Motive

The primary scientific motivation behind ANTARA is the need for improved understanding of the trapped proton environment within the South Atlantic Anomaly and its implications for spacecraft operations, radiation modelling, and near-Earth space science.

The South Atlantic Anomaly represents the region where the inner Van Allen radiation belt approaches closest to Earth due to the offset and tilt of Earth's magnetic dipole. Spacecraft traversing this region are exposed to significantly elevated fluxes of energetic charged particles, particularly high-energy protons capable of producing Single Event Upsets (SEUs), sensor noise, detector saturation, memory corruption, semiconductor degradation, and cumulative radiation damage.

ANTARA aims to contribute toward this scientific need by directly measuring Unidirectional Differential Proton Flux (FPDU) within the energy range of 5 MeV to 100 MeV using an onboard proton detection payload. The mission seeks to generate spatially and temporally correlated radiation datasets that can be compared against existing radiation models including AP8 and AP9.

1.2.3 National Motive

From a national perspective, ANTARA aligns strongly with India's growing emphasis on indigenous space technology development, scientific self-reliance, and advanced research capability. The mission contributes toward India's requirement for indigenous expertise in spacecraft systems engineering, radiation-aware electronics, mission operations, and advanced satellite technologies.

Also, the satellites in the past few years have faced various anomalies with this region due to which communication signals with them were lost. Data collected by ANTARA maps the shifting borders of the SAA, allowing Indian satellites to shield or power down domestic defense and communication satellites before they sustain damage. ANTARA provides an opportunity

for low-cost effective and indigenous space tech to be tested via its CubeSat.

1.3 Mission Significance

1.3.1 Primary Flight Objective

The primary objective is to measure and map unidirectional differential proton flux in the South Atlantic Anomaly over the 5–100 MeV range.

1.3.2 Secondary Mission Objectives

The secondary objectives are to examine East–West asymmetry, characterize the bifurcation of the SAA, investigate atmospheric-loss coupling, and develop corrections to radiation-model predictions from ANTARA measurements. Storm-time comparisons remain an exploratory objective dependent on available observations. The detector architecture and channel definition remain under development; the science-success criteria are proposed targets to be tested against actual detection and sampling capability.

1.4 Mission Success Criteria

Mission success criteria define the measurable conditions under which the ANTARA mission can be considered operationally and scientifically successful. These criteria establish a structured framework for evaluating mission performance at different levels of achievement throughout the mission lifecycle. The criteria are divided into three progressive categories:

- Minimum Mission Success (MMSC)
- Full Mission Success (MSC)
- Extended Mission Success (EMSC)

1.4.1 Minimum Mission Success Criteria (MMSC)

The MMSC defines the lowest acceptable bar for declaring the mission a scientific and engineering success. Meeting the MMSC demonstrates that ANTARA has survived launch, commissioned its payload, operated within the SAA environment, and produced scientifically interpretable, calibrated flux data at coarse spatial resolution.

MMSC mission duration targets. Mission duration: 4-6 Months minimum. Orbital count: 1750 orbits (4 months) to 2650 orbits (6 months). SAA transit count: 700 to 1050 SAA transits across the mission duration. Minimum SAA passes for MMSC science: $\geq$ 150 passes with payload operational and telemetry received.

MMSC spatial mapping requirements. Grid resolution: 5° × 5° in geographic latitude and longitude. SAA region: Approved count-rate polygon; vertices in the coverage contract. Geometric polygon-overlap cells: 92; science eligibility requires validation. Cell dwell time per pass: ~74 seconds per 5° latitude band at 7.5 km/s.

MMSC science requirements. The mission shall produce a calibrated 5°×5° flux map covering at least 70% of grid cells within the approved SAA polygon with valid measurements ($\geq$3 independent SAA passes per cell). Coverage shall include the SAA flux core (centred approximately on 310°E, 30°S) to $\geq$90% even if peripheral cells fall below the 70% global threshold.

Each populated spatial cell shall report a calibrated differential proton flux value in units of particles/cm²/s/sr/MeV for each operational energy channel. σ_F/F $\leq$ 50% per cell per channel. Requires approximately N $\geq$ 4 counts per cell per channel. Flux maps shall be produced for a minimum of 2 energy channels independently.

MMSC Scientific Deliverables

- Calibrated 5°×5° differential proton flux map in at least 2 energy channels
- Flux uncertainty map (σ_F/F per cell per channel)
- Pass-count coverage map (N_passes per spatial cell)
- Detector performance report: channel stability, livetime fractions, dead-time characterisation
- model-baseline comparison plots with residual map at 5° resolution
- Mission timeline with all anomaly flags and excluded data intervals documented

1.4.2 Mission Success Criteria (MSC)

MSC mission duration targets. Required duration: 12 months - 14 months minimum. Orbital count: 5400 orbits (12 months) to 6300 orbits (14 months). SAA transit count: 2150 to 2500 SAA transits. Minimum SAA passes for MSC science: $\geq$ 900 passes with payload operational.

MSC spatial mapping requirements. Grid resolution: 1° × 1° in geographic latitude and longitude. Geometric polygon-overlap cells: 2,020; science eligibility requires validation. Cell dwell time per pass: 15 seconds per 1° latitude band at 7.5 km/s.

MSC science requirements. The mission shall produce a calibrated 1°×1° flux map covering at least 90% of approved SAA polygon cells that contain detectable flux. Each populated 1°×1° cell shall achieve σ_F/F $\leq$ 25% in at least the two lowest-energy operational channels. Requires N $\geq$ 16 counts per cell per channel. Independent flux maps for all operational energy channels, co-registered on the same 1°×1° grid, released as a unified multi-band atlas. Minimum of 5 independent SAA passes per populated cell for inclusion in primary science map. Monthly-averaged sub-maps produced and compared. Month-to-month variation exceeding 3σ in any cell shall be flagged.

MSC Scientific Deliverables

- 1°×1° multi-channel calibrated differential proton flux atlas (primary science product)
- Per-channel uncertainty maps at 1° resolution
- radiation-model residual maps per energy channel
- East-west asymmetry map with statistical significance per cell
- Monthly sub-maps (12–14 monthly averages) for temporal trend analysis
- Full public data release: calibrated counts, flux products, metadata, and uncertainty estimates

1.4.3 Extended Mission Success Criteria (EMSC)

EMSC mission duration targets. Required duration: 18 months minimum - 20 months nominal. Orbital count: 8100 orbits (18 months) to 9000 orbits (20 months). SAA transit

count: 3200 to 3600 SAA transits. Minimum SAA passes for EMSC science: $\geq$ 2400 passes with payload operational.

EMSC spatial mapping requirements. Grid resolution: 0.5° × 0.5° in geographic latitude and longitude. Geometric polygon-overlap cells: 7,940; science eligibility requires validation. Cell dwell time per pass: ~7 seconds per 0.5° latitude band at orbital velocity.

EMSC science requirements. The mission shall produce a calibrated 0.5°×0.5° flux map. Primary science map includes all cells where σ_F/F $\leq$ 15%. $\geq$95% coverage of cells where AP9 predicts flux exceeding 1% of the mission peak flux. σ_F/F $\leq$ 10% in at least one energy channel per cell, and $\leq$ 15% in all other operational channels. Minimum of 5 independent SAA passes per primary-map cell. 0.5° map produced for each operational energy channel independently; minimum 3 channels. Final 0.5° science map produced as a time-integrated product over full EMSC window, plus two 9–10 month sub-period maps.

Table 1: Comparison of success tiers

Parameter	MMSC	MSC	EMSC
Mission Duration	4–6 months	12–14 months	18–20 months
Min SAA Passes	$\geq$ 150	$\geq$ 900	$\geq$ 2400
Spatial Resolution	5° × 5°	1° × 1°	0.5° × 0.5°
Geometric overlap cells	92	2,020	7,940
SAA Coverage	$\geq$ 70% ($\geq$90% core)	$\geq$ 90%	$\geq$ 95% (primary)
σ_F/F Threshold	$\leq$ 50%	$\leq$ 25%	$\leq$ 10–15%
Min Passes/Cell	$\geq$ 3	$\geq$ 5	$\geq$ 5
Energy Channels	$\geq$ 2	$\geq$ 4 (all)	3 sustained
AP9 Comparison	Basic ratio check	Full residual map (1°)	Sub-map residuals (0.5°)

1.5 Scientific Interpretation of Feasibility Evidence

Science assessment. MMSC, MSC and EMSC are retained as committed requirements. Grid-cell occupancy, detector acceptance, energy migration and systematic uncertainty must be coupled with the chosen payload response before compliance can be claimed.

Coverage acceptance contract. The polygon is the supplied count-rate domain, using east-positive longitude in $[-180, 180)$ and positive-area cell overlap. Its geometric counts are not automatic science-success denominators. The coverage, pass, channel, precision and temporal requirements above remain committed. Eligible flux/core definitions and independent observation rules require acceptance before calculating compliance. Repeated samples in one visit are not independent passes. Counts require science-enabled exposure, qualified response/background and usable stored/downlinked products; pure Poisson thresholds alone do not establish calibrated flux precision.

2 Mission Overview and Orbit Design

The ANTARA mission has been conceived as a dedicated scientific CubeSat mission focused on characterizing the high-energy proton radiation environment within the South Atlantic Anomaly (SAA). The spacecraft is intended to operate in a near-circular Sun-Synchronous Low Earth Orbit. The detailed orbit propagation and trade-study results are provided in Appendix A.6 and Appendix B.

2.1 Spacecraft Overview

ANTARA is a 3U CubeSat with an on-board computer, an electrical power system, an attitude determination and control system, a communications system, and the scientific payload. The initial orbit is approximately 600 km at an inclination of 97.6°. No propulsion system is planned.

The 3U spacecraft dimensions are approximately $100 \times 100 \times 340.5$ mm. The mass budget and 3 kg design limit are assessed in Section 5.3.

2.2 Orbit Selection

The preliminary orbit baseline is a 600 km Sun-Synchronous Orbit with an inclination of 97.6° and an eccentricity of approximately 0.001. The latest seven-case RAAN power study reports its largest modeled mean generation at 180°; power and battery behavior depend on the insertion RAAN. ANTARA does not control the launch RAAN, so the spacecraft is being assessed across multiple RAAN cases. The approved future analysis epoch is 1 January 2029; the evaluated studies retain the source epochs and assumptions identified in Section 4.

Table 2: Preliminary orbit parameters

Parameter	Mission baseline / current status
Initial altitude	Approximately 600 km
Inclination	97.6°
Eccentricity	Approximately 0.001
RAAN	Launch-dependent; 0° and 30° are favorable cases, not requirements
Propulsion	None

The mission targets approximately five years of operational science. The passive-decay study and its propagation endpoint are assessed separately in Section 5.5.

2.3 Mission Lifecycle

The mission is divided into Launch, Deployment, Cold and Dark, Detumbling, Commissioning, Nominal Operations, Extended Operations, and Mission End phases. Following deployment the spacecraft activates after the planned delay, enters Cold and Dark, performs detumbling, and proceeds toward power-positive operations and commissioning. Nominal operations consist of Sun Pointing, Detection, and Relay, with Safe Mode for recovery.

The extended phase begins after nominal mission objectives have been met. The minimum, full, and extended mission duration targets are defined in Section 1.4. The detailed phase descriptions and mode transitions are retained in Appendices B and A.1.

3 Spacecraft Architecture and Ground Segment

The ANTARA spacecraft is a 3U CubeSat with the on-board computer as the main command and operations unit. The subsystem design currently uses preliminary COTS components along with the in-house scientific payload. Detailed specifications, operating estimates, and the original component tables are provided in Appendix C.

Feasibility assessment convention. SUPPORTED BY SIMULATION identifies a result demonstrated within the stated computational model, including analytical and reference-software calculations. CONDITIONALLY SUPPORTED identifies partial evidence or an application dependent on stated assumptions. NOT YET ASSESSED identifies missing performance analysis; NOT SUPPORTED BY CURRENT RESULTS identifies a tested shortfall. These assessments preserve the mission-success requirements in Section 1.4.

3.1 Electrical Power System (EPS)

The spacecraft uses one fixed solar panel on the +Z face and two panels stowed on the -Y face that deploy toward the +Z direction. The mission power studies use a 25 Wh battery planning capacity. The proposed ISISPACE ICEPS2 Type A has a manufacturer-rated 28.8 Wh battery-pack configuration; the mission simulation is assessed at 25 Wh. The original RAAN trade study reports average generated power of 11.05 W and 11.40 W for 0° and 30° RAAN respectively. These end-of-life study cases support array sizing across illumination conditions. In the corrected annual DITL, recorded generation is 50,288.809 Wh against 30,866.103 Wh of modeled demand, and minimum recorded battery energy is 15.098 Wh. The evaluated schedule therefore maintains positive battery energy while serving its modeled loads. The DITL source context is defined in Section 4.1; its inputs are not combined with the separate RAAN cases to infer a common EOL margin. Cell operating limits, startup reserve and peak bus demand remain the constraints on applying this result to the selected hardware.

Operating constraints and evidence. The EPS must sustain the scheduled mission loads with the 25 Wh planning battery across representative launch-dependent RAAN and eclipse cases, without an accumulating energy deficit. Annual solar/load integrals: 50288.809/30866.103 Wh. Generation exceeds modeled demand; application to the selected EOL configuration is conditional. The EPS must keep battery energy within selected-cell operating and depth-of-discharge limits during startup, science, Relay and Safe Mode. Numerical SOC/DoD limits remain open allocations. Corrected annual minimum 15.097556 Wh; actual capacity and supplier limits unknown.

The EPS must supply simultaneous and transient load demand within bus voltage, current and ripple allocations, including payload acquisition, radio transmit and actuator peaks. These limits remain open allocations. Mode loads are represented; peak current and bus voltage/ripple allocations have not been assessed. The EPS must generate the allocated Sun-Pointing energy using the fixed +Z panel and two deployed panels, with illumination, pointing and conversion losses explicitly accounted for. Configuration defined; generating and deployed-area identities differ.

The EPS must reserve energy for detumble, safe-state survival and recovery until the next usable generation opportunity; maximum recovery duration and energy reserve remain open allocations. One 2460 s detumble event; left-held load integral 1.528000 Wh; reserve not allocated. The EPS must maintain the mission energy allocation through the approximately five-year operations objective using traceable EOL degradation and cycling assumptions. EOL

study inputs exist; average generation is not an EOL margin.

Feasibility assessment. The corrected annual scenario sustains the modeled load schedule with positive recorded battery energy throughout. This supports operational energy feasibility for that scenario. Applying the result to the selected array and battery remains conditional on common load, degradation, reserve and cell-limit assumptions.

3.2 Communications System (COMMS)

The preliminary communications system uses the GomSpace NanoCom AX100 and directional and omnidirectional spacecraft antennas. The planned primary ground station is at Goa; a crossed-Yagi ground antenna is being considered. The corrected contact study supplies the visibility windows used for capacity calculations; its elevation mask is unspecified, so no additional mask is imposed. A secondary station is an option if the primary station cannot support the required operations.

The communications model allocates the full ideal gross rate of 19.2 kbps, or 2400 bytes/s, to science during eligible Relay contact. Annual eligible capacity is 428,836,800 bytes, accommodating the evaluated 46,637,364-byte workload with zero final backlog. This is a modeled service opportunity; actual net throughput depends on the RF link and protocol. The 256 KiB dedicated science queue and 128 MiB persistent allocation are evaluated separately in the OBC assessment and Appendix A.5.5.

Operating constraints and evidence. The communications planning budget must use 19200 bit/s (2400 bytes/s), 100% science allocation and no arbitrary derating during eligible visibility-intersect-Relay windows. Corrected annual Relay opportunity 428836800 bytes; full-rate arithmetic reproduced. The communications schedule must return the generated science workload within the accepted retention/outage envelope, with bounded backlog and no unaccounted science loss. The sustained workload/envelope remain open allocations. Corrected annual EMSC slot sensitivity:46637364 bytes; nominal peak75336 bytes; response/timing open.

The communications interface must support BITS Goa with a traceable station, frequency/modem and elevation-mask configuration; capacity must use the producing mask exactly once. Contact study identifies GS1_India; elevation-mask and RAAN inputs are unspecified. The communications protocol must detect corrupted or missing records and resume interrupted transfers with identifiable sequence, acknowledgement and duplicate handling. v3 CRC/sequence tested; packet reception and retry protocol absent.

The communications design must close RF link and net science-throughput budgets for the selected hardware and operational geometry before claiming physical data-return compliance. RF margin and net application throughput have not been calculated for the selected link.

Feasibility assessment. The eligible Relay capacity accommodates the sampled annual histogram workload at the modeled gross rate. Communications feasibility is conditional on the selected RF link delivering the required service and on the accepted outage envelope.

3.3 Scientific Payload

The current FPDU concept uses five $6\times6\times14$ mm main scintillator blocks in series along X, plus one $6 \times 6 \times 2$ mm front scintillator. Particles enter the $+X$ face and travel inward toward $-X$. Main-block SiPMs attach to $\pm Y$ (ten sensors), with one further front SiPM: eleven total. Teflon

coats $\pm Z$ at 0.11 mm per face, giving 6.22 mm coated Z dimension. EJ-550 grease provides optical coupling. These dimensions define the current design; material selection, inter-block optical isolation and packaged CAD remain to be established. The historical silicon entrance is removed; a rear scintillator is optional and unselected. The proton-energy requirement remains 5–100 MeV.

The proposed chain is scintillator, SiPM, ASIC, ADC, MCU or FPGA, classification, reconstruction, 64-bin histogram, and storage/downlink. ASIC shaping is 20 ns. Coincidence-window references are 200 ns nominal, at least 250 ns including allowance, and approximately 300 ns preferred. A separate stated 20 ns post-shaping interval has unresolved meaning. End-to-end dead time includes ADC and processing latency, which are not yet allocated. MCU is preferred only if the entire architecture meets timing, throughput, buffering and power needs; no processor is selected.

The incident-rate study predicts proton maxima of 358.920 Hz (mean map) and 975.659 Hz (99th-percentile map), using isotropic conversion, $G = 1$ cm^2 sr, unit efficiency and an integral threshold of 5.5 MeV. These 2026-01-01, 600 km maps are preliminary references for 2029 planning. They provide a spatial rate basis for electronics sizing; electron incidence must also be included. The assumed geometric factor and efficiency are not demonstrated detector acceptance, and the calculation does not model shielding or energy reconstruction. The superseded 400 Hz estimate and its assumptions are retained in Appendix A.2.

Routine science uses the 292-byte Science Record v3: entry/exit timestamps, geographic position, 64 uint32 histogram bins, sequence, combined mode/health marker and CRC. Prefer three nonoverlapping readings per eligible cell visit, with fewer when actual acquisition timing permits; valid zero-count intervals are retained. No arbitrary minimum duration is imposed. Record validity, accumulated statistical precision and mission-tier success are separate. Table 10 defines the reference encoding; radio framing is excluded.

Operating constraints and evidence. The payload must provide calibrated differential trapped-proton flux over 5–100 MeV, with energy-dependent response, background discrimination and uncertainty adequate for the approved tier products. No calibrated six-scintillator response; 5.4/5.5 MeV study cutoffs do not close 5 MeV. The payload science chain must enable a 5°×5° MMSC map: at least 70% polygon coverage, 90% core coverage, 3 independent passes/cell, at least 2 channels and fractional flux uncertainty $\leq$50% per populated cell/channel. 92 geometric cells; eligible/core and independent-pass definitions unresolved.

The payload science chain must enable the MSC 1°×1° atlas: at least 90% detectable-cell coverage, 5 independent passes/cell, uncertainty $\leq$25% in the two lowest operational channels, all operational channels, and monthly maps with >3σ changes flagged. Detailed criteria say all operational channels; summary specifies $\geq$4. Both retained pending reconciliation. The payload science chain must enable 0.5°×0.5° EMSC products: $\geq$95% of cells above AP9 1%-of-mission-peak eligibility, 5 independent passes/cell, $\geq$3 channels, $\leq$10% uncertainty in one channel and $\leq$15% in every other operational channel. Primary-map $\leq$15% rule and two 9–10 month submaps remain binding; required-G percentiles are not compliance.

The payload mechanical/readout interface must conform to five longitudinal 6×6×14 mm blocks, a 6×6×2 mm front element and eleven SiPMs, with no silicon entrance stage. Geometry arithmetic/configuration checked; package/material/optics unselected. The payload timing interface must distinguish 20 ns ASIC shaping from the 200 ns coincidence target and $\geq$250 ns allowance (about 300 ns preferred); it must not assign these values as detector dead time. 20 ns shaping and coincidence timing are separate inputs; end-to-end latency and dead

time are undetermined.

The payload readout must handle the declared mixed-species peak/burst trigger envelope without unflagged saturation, pileup, loss or counter overflow. Rate, loss and dead-time allocations remain open allocations. Proton-only400 Hz is not a bound; electron incident proxies can dominate. The payload response allocation must provide energy-dependent effective geometric factor and livetime sufficient for tier exposure/precision, including background and reconstruction uncertainty. Required acceptance values remain conditional on verified exposure. Broad-band G95≈0.36–0.50 cm² sr at10% is ideal counting arithmetic, not achieved G.

The payload science accounting must conserve counts across acquisition time and effective energy-bin boundaries, retain valid zero-count exposure, and carry livetime, background and calibration identity into accumulated cell/channel uncertainty. Reference conservation tests pass; legacy expected counts are not reconstructed events. The payload reference science format must define entry/exit timestamps, latitude/longitude/altitude, 64 provisional uint32 bins, sequence, 16-bit mode/health marker and CRC in 292 bytes excluding radio framing. Codec/schema and CRC/count tests support the logical interface.

The payload scheduler must prefer 3 readings per eligible visit, seek 2 when feasible and retain 1 when necessary, with nonoverlapping valid intervals, correct cell assignment and conserved counts/exposure; no arbitrary minimum duration or count is imposed. Reference partitions tested; exact crossings and hardware boundaries unknown.

Feasibility assessment. The scintillator geometry, distributed SiPM readout and compact histogram interface define a coherent instrument concept. Incident-rate and acceptance calculations support sizing studies. Scientific feasibility remains conditional on energy-dependent detector response, mixed-species handling, effective acceptance and reconstruction uncertainty. Section 1 retains all tier criteria, including the MSC channel-count discrepancy requiring resolution.

3.4 Attitude Determination and Control System (ADCS)

ANTARA intends to use an integrated reaction-wheel and magnetorquer attitude-control system; the earlier CMG description is superseded. The supplied TensorTech ADCS-10m databook describes a CMG-equipped unit and therefore does not confirm the intended reaction-wheel plus magnetorquer architecture. The ADCS vendor/product selection remains open. The main attitude functions are initial detumbling, Sun Pointing, payload +X pointing during SAA operations, +Y relay pointing toward the ground station, and Safe Mode recovery. The applicable pointing and actuator-authority envelopes depend on the selected unit and payload/antenna acceptance. Mode transitions are given in Appendix A.1.

The supplied historical detumble summary contains 50 000 trials, with 16 133 reported successes (32.266%). Successful trials alone have median completion 951.5 s and 95th percentile 2569.0 s. The 33 867 nonconverged rows retain infinite completion values; their censor times and generating configuration are missing. All trials contain independently specified nonzero initial accelerations, which the current source rejects. This synthetic campaign is incompatible with the current initial-condition model and cannot establish detumble reliability. The integrated scenario separately completes a 2460 s detumble event and proceeds to scheduled operations, supporting startup sequencing for that evaluated case. Detailed interpretation is in Appendix F.7.

Operating constraints and evidence. The ADCS must detumble and acquire a safe Sun-Pointing state from the accepted deployment attitude/rate envelope within the EPS startup allocation; envelope, deadline and reliability target remain open allocations. One 2460 s event; historical 16133/50000 successes are source-incompatible. The ADCS must maintain payload +X science and +Y Relay boresights within allocated acceptance/link pointing-error envelopes during valid acquisition/transfer. Numeric error and jitter allocations remain open allocations. No geometry-matched acceptance or selected-antenna error allocation.

The ADCS must hold Sun Pointing and perform recovery/Safe Mode attitudes within the EPS generation and thermal allocations. Numeric attitude limits remain open allocations. Mode concept defined; common pointing-energy/thermal limits not closed. The ADCS must complete and settle science/Relay slews before their qualified windows and keep transition overhead within the mission exposure/data-return allocation. Deadline and settling limits remain open allocations. Complete day/year timestamp spans; telemetry/transition/slew duration definitions disagree.

The ADCS must keep wheel momentum and torque within selected actuator limits and unload using magnetorquers against the declared disturbance envelope. Numerical authority/margin allocations remain open allocations. Selected actuator/field coupling is undetermined; archived zero wheel momentum does not establish unloading capability. The ADCS must report acquisition, saturation and pointing faults to the mode controller and enforce safe guards before declaring science/Relay readiness. FSM exists; historical boundary/SAFE timing discrepancies remain.

Feasibility assessment. The integrated scenario demonstrates a startup-to-operations sequence with time allocated to attitude transitions, science and Relay. Applicability to the intended reaction-wheel/magnetorquer system remains conditional on actuator authority, deployment conditions and pointing allocations. The incompatible Monte Carlo campaign cannot establish startup reliability.

3.5 On-Board Computer (OBC)

The source design uses the GomSpace NanoMind A3200 as the preliminary on-board computer. The OBC is responsible for mission sequencing, spacecraft state monitoring, command execution, payload control, data handling, and communications scheduling. The source document gives approximately 0.17 W as the average OBC power consumption for its stated operating conditions. Reference data handling uses a 292-byte record and an illustrative 804-byte assembly working set. The corrected annual workload reaches a 75,336-byte nominal queue, fitting the candidate 256 KiB science allocation before metadata. A seven-day outage increases the queue to 1,010,028 bytes and requires nonvolatile spill; the candidate 128 MiB store accommodates that case before reserves. These calculations support the science-buffer architecture; total OBC execution, memory and fault-recovery resources require separate allocation.

Operating constraints and evidence. The OBC must allocate record assembly separately from the science queue: 292 bytes per packed reference record and explicitly budgeted acquisition banks, stack and DMA; a two-bank histogram fixture adds 512 bytes. 804-byte illustrative payload working set; not total RAM or selected architecture. The OBC science queue must retain the corrected nominal workload without loss and provide nonvolatile spill for the accepted outage envelope; science RAM, persistent retention and total system memory must be allocated separately. Corrected annual nominal EMSC peak75336 bytes; seven-day

outage1010028 bytes exceeds256KiB RAM.

The OBC persistent science store must retain all unacknowledged records through the accepted contact-outage/retention envelope, including metadata/reserves; that envelope remains an open allocation. Corrected annual seven-day EMSC outage peak1010028 bytes;128MiB exceeds it before reserves. The OBC must provision total FSW/system memory and execution capacity separately from record assembly, queue and persistent storage, covering control, reconstruction, commands and fault handling. Limits remain open allocations. No integrated worst-case memory/execution budget.

The OBC must maintain record integrity and sequence continuity across interrupted writes, reset and Relay interruption, recovering only complete records and retaining unacknowledged data. Reference CRC passes; atomic media writes and acknowledgements not implemented. The OBC must execute approved state-machine guards, validate commands, report faults and enter a defined safe state without uncommanded detector/Relay operation. The state-machine concept defines control and fault sequencing; command-integrity implementation is not assessed.

The OBC must preserve payload timestamps, cell/configuration identity and linked live-time/background/quality metadata through storage and ground reconstruction, without treating unknown health as nominal. v3 has linked-session metadata; size/cadence and clock performance open.

Feasibility assessment. The calculated reference workload fits the candidate nominal science queue. Seven-day outage recovery requires persistent spill, which the candidate store accommodates before reserves. The separate 187498164-byte no-downlink grid sensitivity exceeds 128 MiB and is retained as a distinct retention case. Total execution and memory allocations remain to be established.

3.6 Spacecraft Coordinate System

The spacecraft body frame uses +X as the RAM face and payload boresight, +Y as the directional antenna face, +Z as the fixed solar-panel face, -Y as the deployable-panel packed face, -Z as the omnidirectional communications antenna face, and -X as the deployment face.

Operating constraints and evidence. The subsystem interface definitions must preserve +X payload entrance with inward −X particle travel, ±Y main SiPMs, ±Z reflector faces, +Y relay boresight and +Z fixed-array orientation. Body-frame configuration and diagrams agree, providing a consistent geometric interface. The navigation/time interface must bound clock and position errors so each science interval is unambiguously assigned to its cell/exposure window. Numerical errors and synchronization allocations remain open allocations. Integer scaling and POSIX convention defined; actual accuracy unknown.

The integration ICD must allocate power, harness/connector, data, thermal and safe-state interfaces with explicit ownership and compatible command/status semantics. No closed common hardware/ICD baseline.

Feasibility assessment. The common body frame gives a consistent geometric interface for payload, antenna and array. Integrated performance remains conditional on alignment, time/position accuracy and compatible electrical and command interfaces.

3.7 Thermal and Structural Configuration

The supplied seven-node thermal history spans 24 hours of recorded labels. Independently reproduced internal-node extrema are -14.061 to 21.159°C; external-face extrema span -25.958 to 31.239°C. The seven-node model uses an independent synthetic heat schedule. Its temperature envelope provides a quantitative basis for component selection and thermal design. Component thermal compliance remains unassessed because node-to-component limits are unavailable; the reported hot/cold extrema are from one history, and the $\pm 10^{\circ}$C allowance is not a component margin. DITL-coupled bounding cases are a remaining concept assessment. Appendix A.4 presents the thermal model and preliminary 3U mechanical arrangement. The specified-mass subtotal is 1536.3 g; payload, deployment and panel-quantity allocations are needed to assess the complete 3 kg design.

Operating constraints and evidence. The thermal subsystem must keep each selected component within traceable operating and survival temperature limits for nominal and Safe Mode cases; numerical limits/node mapping remain open allocations. One history: internal -14.061 to21.159°C; no component compliance limits. The thermal assessment must bound hot, cold and eclipse temperatures using mission-compatible attitude and mode-dependent internal heat loads, including startup, science, Relay and safe recovery. Supplied extrema are one history; 1.9–4.9 W schedule is independent of DITL.

The thermal interfaces must allocate battery charge/discharge temperature constraints and payload temperature-dependent calibration/stability limits. Both numerical allocations remain open allocations. No selected-part limits or payload thermal response characterization. The thermal-mechanical design must keep gradients and resulting distortions/stresses within allocated alignment and material limits; numerical limits remain open allocations. 31.704°C face spread is a node difference, not structural stress.

The structure/integration budget must fit the selected 3 U deployer envelope and keep the complete design mass, including all allocations and contingency, at or below 3.0 kg. 1536.3 g is incomplete BOM; provider ICD, payload and deployment allocations open. The structure must withstand applicable launch loads and meet provider modal, vibration and shock requirements without loss of function or interface integrity. Numerical load/frequency limits remain open allocations. Launch-provider load envelope and structural/modal analysis are not available.

The solar deployment/retention mechanism must retain both panels safely through launch and release/verify deployment without collision, harness damage or loss of the required generation geometry. Deployment architecture is defined; mechanism loads, clearances and retention life are not assessed. The mechanical integration must preserve FPDU optical/readout interfaces and body-frame clearances through assembly, thermal cycling and operational life. Active material dimensions exclude coating, gaps, sensors and housing.

Feasibility assessment. The thermal model predicts a finite temperature envelope for its evaluated heat schedule, providing a basis for component selection and thermal design. Component compliance and structural feasibility are not yet assessed against selected-part limits, a complete mass allocation and launch-provider loads.

3.8 Ground Segment and Mission Operations

The primary ground station is planned at the Goa campus, approximately 15°N, 74°E. A second station is a contingency only if the primary station cannot meet operational requirements.

The ground segment provides tracking, telemetry reception, commanding, and science-data retrieval. The operational sequence uses Cold and Dark, Detumble, Sun Pointing, Detection, Relay, and Safe Mode, with Slewing as an attitude-transition state in the simulation. The full state-machine logic is given in Appendix A.1.

Operating constraints and evidence. The ground/operations chain must plan and verify the approved observation windows: MMSC4–6 months and≥150 operational received passes; MSC12–14 months and≥900 operational passes; EMSC18–20 months and≥2400 operational passes. A complete annual operations timeline is available; accepted detector observations and tier pass counts are not yet assessed. Ground science processing must reconstruct versioned cell/channel products from received records and linked exposure/calibration metadata, retaining zero counts, excluded-data reasons and independent-passage identifiers. Reference record codec defines the ground interface; calibrated map reconstruction is not yet assessed.

Operations verification must use complete, time-aligned telemetry, contact and transfer records for every claimed mission-window data-return result. Corrected telemetry spans365days; contact/native data totals reconcile; mode/slew semantics unresolved. Ground operations must establish primary-station availability and a missed-contact recovery schedule that maintains the accepted science retention envelope, including command/fault response. Availability and outage limits remain open allocations. Corrected longest intercontact gap48458s; actual station outage commitment unknown.

Feasibility assessment. The annual scenario demonstrates repeated acquisition and Relay opportunities over a complete operations year. The resulting service schedule supports the sampled data workload. Mission-tier delivery remains conditional on detector response, independent-pass accounting and station availability.

3.9 System Integration

The subsystem analyses support the resource and operational conclusions within their stated scenarios. Section 5.1 assembles these findings without combining incompatible configurations.

Operating constraints and evidence. The system feasibility assessment must identify a common mass, epoch, orbit/RAAN, duty-cycle, response and resource baseline for each combined margin and disclose incompatible evidence rather than pool it. Studies retain distinct epoch, mass and detector-response assumptions; no combined margin is inferred.

Feasibility assessment. The evidence supports scenario-level operations and resource feasibility. Each study retains its own configuration; the integrated mission conclusion in Section 5 distinguishes demonstrated results, conditional applications and unassessed constraints.

4 Integrated Mission Feasibility

The ANTARA simulation work combines orbit motion, power generation and consumption, attitude control, mode transitions, SAA operations, and ground-station contacts. The corrected DITL simulations demonstrate how these activities share time and energy over startup, a nominal day and a complete operations year. The detailed simulation descriptions, tables, and plots are retained in Appendices D and A.3.

4.1 Day-in-the-Life Operations

Simulation context and assumptions. The corrected LEOP 24-hour, No-LEOP 24-hour and LEOP 365-day studies form the operational evidence used here. Their contact and SAA timestamps identify an epoch of 1 March 2000 UTC; the approved 1 January 2029 mission configuration was not simulated by these runs. Each scenario is assessed separately. Producing configuration records do not specify RAAN, mass, battery capacity, integrator, hardware selection or elevation mask. The day cases share recorded positions and contact windows, but their comparison is a scenario sensitivity rather than an isolated causal test of LEOP.

Telemetry occupancy and energy integrals hold each state over $[t_i, t_{i+1})$ and exclude terminal-row duration. Transition and slew-event logs use differing duration definitions, detailed in Appendix F. Battery percentages use the 25 Wh planning reference. Acquisition gates and expected counts use the study's legacy response; they describe observing opportunity, not accepted events from the current FPDU. These assumptions apply to all DITL results in this section.

Table 3: Integrated DITL results using left-held telemetry intervals

Metric	LEOP 24 h	No-LEOP 24 h	LEOP 365 d
Recorded timestamp span (s)	86,400	86,400	31,536,000
Cold and Dark (s)	122	0	122
Detumble (s)	2,460	0	2,460
Sun Pointing (s)	79,268	81,852	29,660,918
Detection (s)	2,676	2,674	894,054
Relay (s)	894	894	269,876
Slewing (s)	980	980	375,688
Safe (s)	0	0	332,882
Geographic SAA (s)	2,696	2,696	881,882
Acquisition gate (s)	758	758	247,258
Minimum recorded battery (Wh)	15.128	18.004	15.098
Recorded solar integral (Wh)	145.575	150.403	50,288.809
Recorded load integral (Wh)	85.167	85.719	30,866.103
Native generation (source MB)	0.009	0.009	3.514
Native transfer (source MB)	0.009	0.009	3.514

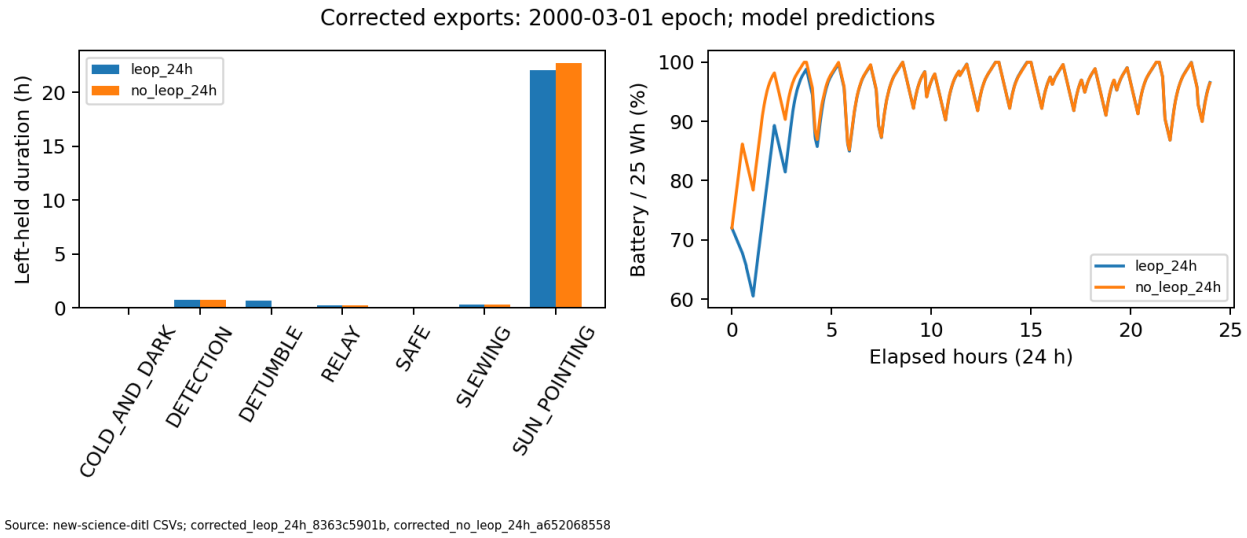


Figure 2: Day-run mode occupancy and battery energy normalized to the 25 Wh planning reference.

Startup and science acquisition. Both day runs record 2,696 s of geographic SAA dwell and 758 s of acquisition gate. The LEOP case completes one 2,460 s detumble event, followed by science and Relay operations within the first day. Removing startup does not increase the sampled acquisition total in this pair. The simulation therefore supports the evaluated startup sequence without identifying startup as the limiting factor for that day's acquisition opportunity.

Annual operational continuity. The annual timeline allocates 29,660,918 s to Sun Pointing, 894,054 s to Detection, 269,876 s to Relay and 375,688 s to Slewing, while recording 332,882 s in Safe Mode. It demonstrates a full-year schedule containing science, communications and recovery states. Safe Mode occupancy is a real operational interruption; it is retained in the resource and exposure accounting. The acquisition gate totals 247,258 s against 881,882 s geographic SAA dwell, or 28.038%. This establishes usable modeled acquisition opportunity, with geographic dwell and Detection occupancy distinguished from actual gated exposure. Independent passes and calibrated science-tier success require the detector-response assessment in Section 3.3.

Power behavior. Both day cases and the annual case retain positive recorded battery energy, with minima of 15.128, 18.004 and 15.098 Wh respectively. Annual recorded generation of 50,288.809 Wh exceeds the modeled load integral of 30,866.103 Wh. Together with the battery trajectory, these results support sustained operation of the evaluated load schedule. The energy-integral difference is not a stored-energy or component margin; charging limits, losses and selected-cell limits must be represented before interpreting it as such.

4.2 Ground Station and Data Return

The corrected annual contact log contains 820 intervals totaling 244,016 s. Mean daily contact is 668.537 s and minimum daily contact 514 s; the longest intercontact gap is 48,458 s. Visibility alone corresponds to 585,638,400 bytes at 19.2 kbps. Intersecting these contacts with telemetry-labelled Relay yields 178,682 s, or 428,836,800 ideal bytes. The transition-log calculation also gives 428,836,800 bytes of eligible capacity, despite differing mode-duration definitions. The recurring opportunities support scheduled ground relay for this evaluated orbit.

Native annual transfer credit is 3.514083862 source-labelled MB, separate from hypothetical v3 delivery. The source-labelled MB units are retained without assuming a byte conversion. The v3 preferred EMSC workload generates 46,637,364 bytes with peak modeled queue 75,336 bytes and final backlog 0 bytes. It uses corrected, time-aligned sampled gates and Relay visibility at the full ideal 2400 bytes/s; it is not emitted v3 telemetry. Appendix A.5.5 gives tier and outage sensitivity. The calculated service accommodates the sampled annual workload and returns the queue to zero. This supports the scheduled data-return concept under the full ideal gross-rate assumption. RF margin and actual net throughput remain separate from this capacity result. The seven-day outage case requires 1,010,028 bytes of buffering, establishing the need for persistent spill beyond the candidate 256 KiB RAM queue.

4.3 Updated RAAN Trade and Long-Duration Study

The latest supplied RAAN campaign evaluates seven insertion cases from 0° to 180° in 30° increments, with two runs per case. Average generated power varies from approximately 9.02 W (120°) to 11.17 W (180°), and tabulated mean of run-minimum battery SOC varies from approximately 54.4% to 74.7%. These are *study-case outputs*, not a statement that the spacecraft can choose its insertion RAAN. Orbit classification totals are pooled across the two long-duration runs; they must not be mistaken for a one-year orbit tally. The full results, raw-count interpretation and drag-study limitations appear in Appendix H.

The drag CSV gives modeled termination times of roughly 8.2–10.0 years, generally near 300 km altitude. One case ends at 334.93 km, and none models final atmospheric reentry. The five-year *science/operational objective* is therefore distinct from the provisional passive-decay estimate. This separate campaign retains its own configuration and unspecified epoch; its outputs are not combined with the DITL to represent the approved 1 January 2029, 3 kg baseline.

4.4 Operational Feasibility Finding

The integrated scenario demonstrates that the evaluated operational sequence sustains its modeled power demand while allocating time for attitude transitions, science acquisition and ground-station relay. The annual queue calculation additionally supports the sampled workload at the assumed service rate. This is a concept-level operational feasibility result within the source context above. Detector response, selected-actuator pointing performance and component thermal compliance require their own assessments; they are not outputs of the DITL accounting.

5 Feasibility, Risk and Engineering Budgets

5.1 Mission Feasibility

The engineering studies support ANTARA's concept-level operational and resource feasibility for the evaluated scenarios. The corrected integrated simulation maintains positive recorded battery energy while scheduling startup, attitude transitions, science acquisition, ground relay and recovery over a complete year. Its time-aligned data-service calculation accommodates the sampled science-record workload. The mission-level finding is **CONDITIONALLY SUPPORTED**: detector response, selected-actuator capability, component thermal limits and complete mass allocation still determine whether these modeled opportunities can deliver the approved science objectives. Hardware qualification is not a prerequisite for the scenario-level findings established here.

Scientific feasibility — CONDITIONALLY SUPPORTED. The 5–100 MeV objective, spatial mapping tiers and uncertainty criteria remain unchanged. Radiation maps and incident-rate calculations provide a quantitative sizing basis, and the 292-byte histogram record supports compact data return. Ideal geometric factors and unit-efficiency incident rates do not establish accepted detector counts or calibrated flux. Energy-dependent response, background discrimination, effective geometric factor and reconstruction uncertainty remain essential concept questions. MMSC, MSC and EMSC achievement is not yet assessed from accepted science observations.

Orbital feasibility — CONDITIONALLY SUPPORTED. The approximately 600 km, 97.6° orbit studies provide SAA acquisition and ground-contact opportunities. Seven RAAN cases quantify sensitivity to launch-dependent illumination, with average generation spanning approximately 9.02–11.17 W. These results support the orbit concept under the respective study assumptions; insertion RAAN remains uncontrolled, and the studies do not represent a single common 2029 configuration. Passive decay is assessed separately below.

EPS feasibility — CONDITIONALLY SUPPORTED. The corrected annual scenario records 50,288.809 Wh generation, 30,866.103 Wh modeled demand and a minimum battery energy of 15.098 Wh. These results support sustained energy supply for the evaluated schedule. Application to the planned 25 Wh battery requires selected-cell limits, startup reserve, peak electrical demand and consistent array/EOL assumptions; the proposed 28.8 Wh rated pack is not substituted into the calculation.

ADCS feasibility — CONDITIONALLY SUPPORTED. The integrated case completes a 2,460 s detumble event and allocates time to Sun Pointing, Detection, Relay and slews. This supports operational sequencing. The separate synthetic Monte Carlo result of 16,133 successes in 50,000 trials is incompatible with the current initial-condition model and does not establish reliability of the intended reaction-wheel/magnetorquer system. Pointing accuracy, settling time and torque/momentum authority remain unassessed for the selected architecture.

Thermal feasibility — NOT YET ASSESSED against component limits. The seven-node model predicts internal temperatures of -14.061 to 21.159°C and external-face temperatures of -25.958 to 31.239°C for its independent heat schedule. This establishes a preliminary thermal envelope. Component compliance cannot be inferred without operating/survival limits and coupled hot/cold cases; no thermal margin is assigned.

Communications feasibility — CONDITIONALLY SUPPORTED. The annual study provides 820 contacts and 178,682 s of visibility intersecting telemetry-labelled Relay. At the full ideal 19.2 kbps gross rate, calculated capacity is 428,836,800 bytes, accommodating the 46,637,364-byte sampled workload with zero final backlog. This supports scheduled data return; the RF link budget and actual net science throughput remain unassessed for the selected implementation.

OBC and data-handling feasibility — CONDITIONALLY SUPPORTED. The reference codec and queue calculations support compact histogram assembly and storage. The 75,336-byte nominal queue fits the candidate 256 KiB science RAM allocation before metadata. The seven-day outage peak of 1,010,028 bytes is **NOT SUPPORTED BY CURRENT RESULTS** for RAM-only retention; persistent spill is required. The candidate 128 MiB allocation accommodates this modeled outage before reserves. Total OBC memory, execution demand and the accepted retention envelope remain separate allocations.

Mission operations feasibility — SUPPORTED BY SIMULATION for the evaluated schedule. The corrected LEOP 24-hour, No-LEOP 24-hour and LEOP one-year scenarios demonstrate startup-to-operations sequencing, repeated acquisition and ground-relay opportunities, and sustained modeled energy availability. The annual record includes 332,882 s in Safe Mode, so the result includes interruptions rather than assuming continuous science. Actual station availability, accepted detector observations and unresolved transition-duration semantics constrain application beyond the evaluated scenario. The source epoch and assumptions are stated once in Section 4.1.

Mechanical and mass feasibility — NOT YET ASSESSED for the complete spacecraft. The 3U arrangement and defined FPDU geometry establish a packaging concept. The 1536.3 g specified-mass subtotal excludes essential allocations and cannot establish compliance with the 3 kg design limit. Complete payload/deployment mass, panel quantity, packaged clearances and launch-provider loads remain necessary to assess mechanical feasibility.

5.2 Measures of Performance and Effectiveness

Measures of Performance cover battery state, Detection pointing, science-data return, payload livetime and mission lifetime. Measures of Effectiveness cover proton measurements, data return, radiation mapping and model comparison. Appendix E retains the original thresholds and goals; Section 1.4 remains authoritative for science success. Operational opportunity and queue capacity support these objectives without replacing their calibrated-measurement criteria.

5.3 Engineering Budgets

The specified component masses total 1536.3 g against a design target below 3 kg. Payload, deployment hardware and panel-assembly quantity remain incomplete, so no mass margin is reported. Component costs remain TBD. Energy planning uses 25 Wh; the proposed Type A pack is rated 28.8 Wh. Science handling uses the 292-byte reference record and the corrected time-aligned workload, with RAM and persistent storage assessed separately above. Detailed component and budget calculations are retained in Appendices E and A.5.

5.4 Operational and Engineering Risks

Deployment failure, unsuccessful detumbling, loss of pointing, radiation effects, communications outages, battery degradation and thermal excursion remain mission risks. The simulations quantify selected consequences: Safe Mode reduces acquisition opportunity, a seven-day communications outage exceeds the candidate RAM queue, and launch RAAN changes generation. The available risk register does not provide complete likelihood/consequence scores; these effects are retained without assigning unsupported reliability or margin. The detailed register remains in Appendix E.

5.5 End-of-Life and Debris Mitigation

ANTARA has no propulsion and relies on passive orbital decay. The approximately five-year operational science objective is distinct from the modeled 8.2–10.0-year drag-study endpoints across seven RAAN cases and two runs per case. Most traces end near 300 km; one ends near 335 km. These are propagation cutoffs, not atmospheric reentry. Disposal feasibility requires continuation to a physical disposal endpoint with applicable epoch and aerodynamic assumptions. Appendix H retains the study details.

5.6 Remaining Concept-Level Questions

The remaining questions affecting mission feasibility are detector acceptance and reconstruction over the required energy range; mixed-species trigger demand and end-to-end timing; compatible ADCS authority and pointing allocations; selected-cell limits and peak/reserve power allocations; component thermal limits and coupled operating cases; complete mass and mechanical allocations; and an accepted RF service/outage envelope. Science coverage also requires eligible-cell, independent-pass and uncertainty definitions, including resolution of the existing MSC channel-count inconsistency. Fragmented acquisition can exceed the preferred three contiguous readings per visit and requires an explicit record policy. A common configuration and consistent transition timing are needed before combining separate studies into mission-wide margins. These are substantive constraints, not routine qualification prerequisites.

5.7 Further Design Development and PDR Scope

PDR development will refine the RF link budget and net throughput including protocol overhead and retransmission; finalize detector materials, optical response, effective geometric factor and reconstruction; complete mass, structural, deployment and packaged-interface design; define component thermal/environmental qualification; and implement OBC timing, memory, persistent storage, command and subsystem interfaces. These activities build on the demonstrated scenario-level feasibility. The concept questions above remain explicit until resolved by suitable engineering evidence and are not deferred merely as qualification work.

A Detailed Technical Supporting Analysis

A.1 Detailed Operational Mode Transition Logic

The ANTARA concept of operations uses explicit spacecraft modes, bounded transition guards and a safety override. Figures 3 and 4 show the normal mission sequence and supervisory logic. The exact software thresholds are configuration-controlled simulation parameters, rather than independently qualified flight-software settings.

A.1.1 Operational Modes and Attitude Targets

Table 4: Mode functions and commanding intent

Mode	Attitude/control goal	Subsystem behavior and exit condition
Cold & Dark	Deployment initialization; no operational pointing	Minimal EPS/OBC; initialize health monitoring and release startup delay; then Detumble.
Detumble	Reduce body rates using magnetic control	EPS/OBC/ADCS active; exit when rate and stability guard pass; otherwise remain or enter Safe on faults.
Sun Pointing	+Z solar face directed toward Sun	Default power-recovery and power-positive standby; permit scheduled SAA science or contact after safety guards.
Science / Detection	+X payload boresight aligned with required science pointing	Read out FPDU during a permitted SAA opportunity; account for payload warm-up, slew and available energy.
Relay / Downlink	+Y directional antenna aimed toward selected ground station	Radio TX/RX and queued-data transfer during usable elevation window; recover normal pointing after contact.
Passive / Eclipse	Preserve previous attitude during eclipse	Minimize active Sun-search maneuvers; resume Sun pointing or the applicable scheduled activity when illumination and guards permit.
Slewing	Move between required attitude targets	Temporary ADCS phase; next operational mode begins after reorientation/settling; abort on fault or low power.
Safe	Survival, battery conservation and anomaly handling	Suspend nonessential loads; maintain health, available communications and recovery control. Re-enter nominal path only after recovery guards.

A.1.2 Nominal Startup and Science/Relay Sequence

The sequence below extends the observed DITL behavior into a proposed operating procedure. It is an engineering interpretation for planning; the assumptions and their verification limits are stated after Section A.1.3.

Initialization and attitude acquisition. Following release, Cold & Dark provides a protected initialization interval. The proposed sequence checks OBC boot status, time validity, EPS health and the availability of attitude sensors before enabling controlled acquisition. Deployment-related interlocks must be satisfied before an activity requiring deployed hardware is commanded; no additional deployment timing is assumed here. In the corrected LEOP event records, the startup delay is 120 s and detumbling occupies the following 2,460 s.

Their sum, 2,580 s (43 min), is an event-defined reference for reaching the end of detumbling, not the time at which qualified science necessarily begins. The sampled Cold & Dark occupancy is 122 s, so event endpoints and sampled mode durations must not be interchanged.

Detumbling is assumed to terminate on acceptable measured body rates and a stable attitude solution, rather than on expiry of the observed 2,460 s. The spacecraft then acquires Sun pointing, with the +Z solar face directed toward the Sun, to establish a power-recovery state and assess readiness for scheduled tasks. A failed acquisition remains an ADCS/recovery problem; it does not authorize blind continuation to Science. Neither a worst-case startup duration nor a deployment-success probability follows from the supplied LEOP examples.

Preparing and executing science. From Sun Pointing, the scheduler checks an approaching SAA opportunity against predicted slew/settling time, available energy and payload readiness. The intended +X payload pointing is acquired before measurement exposure is credited. A short opportunity may therefore be skipped or used only for its remaining valid interval. Geometric SAA residence alone is insufficient: the acquisition gate also requires instrument availability, valid pointing and acceptable health.

During acquisition, each histogram is attributed to one geographic cell and its own entry/exit timestamps. The preferred three readings per eligible visit use distinct intervals; two or one remain valid when timing or an interruption prevents three. No numerical minimum duration is imposed. A cell crossing, instrument shutdown or loss of acquisition validity closes the current interval. Valid zero-count records are retained, and partitioning a visit must conserve its counts and exposure. Multiple readings in one visit do not constitute independent orbital passages, and record validity does not imply that the cell/channel precision requirement is already satisfied.

Relay and return to standby. A scheduled BITS Goa opportunity permits a slew to the +Y antenna target only when the contact, attitude and resource guards are satisfied. Transfer is restricted to the intersection of visibility and Relay operation. The planning model offers the queued science data the full idealized 19.2 kbps during this intersection, without arbitrary derating; it does not demonstrate net hardware throughput. Contact termination or an override pauses delivery, preserving pending records for a later opportunity. The nominal sequence returns to Sun Pointing after a task, or transitions to another eligible task when its guards permit. In eclipse, retaining the previous attitude is a candidate low-activity fallback only while thermal, energy and attitude constraints allow it.

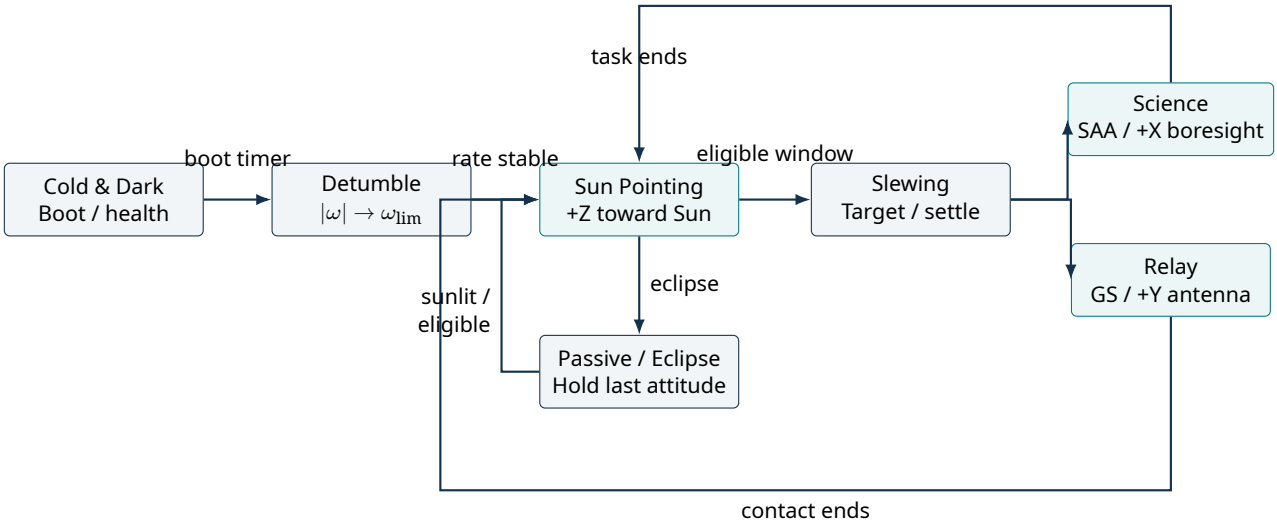


Figure 3: Information-bearing functional flow of the planned mission sequence. Windows and guards come from the mission scheduler. The schematic does not assert that every transition is immediate; slew and settling times are accounted for separately.

A.1.3 Supervisory Priority and Safety Overrides

The supervisor evaluates spacecraft survival before granting any scheduled activity. The proposed policy separates a task that is temporarily ineligible from a condition requiring Safe mode. For example, an unavailable contact or unsettled pointing defers Relay or Science; a critical battery, thermal or control-state violation preempts the active task. The corrected annual run contains Safe-mode occupancy, but this alone does not establish which physical fault caused each entry or qualify the recovery guards.

Priority when opportunities overlap. Survival and recovery take precedence over science and downlink. Among otherwise admissible tasks, this planning interpretation uses the predicted consequences of deferral: loss of a scarce science window, queue overflow before the next contact, and the energy needed to execute and recover from a slew. Relay is not assumed universally superior to Science, nor is simultaneous acquisition and downlink credited. A deterministic scheduler must record the selected task and reason for deferral. Numerical priorities, queue watermarks and energy reserves remain to be allocated; the corrected exposure and backlog results constrain that trade but do not identify an optimal flight scheduler.

Interruption, safe state and recovery. On a critical override, the proposed response closes the valid acquisition interval, marks any uncertain data, inhibits nonessential payload/transmitter loads and preserves recoverable queue and sequence state. Health monitoring and feasible attitude control continue within the surviving power budget. Safe mode is therefore a fault-dependent operating state, not a promise of uninterrupted Sun pointing: excessive rates may first require Detumble, while a sensor fault may prevent the nominal attitude solution.

Recovery requires evidence that the initiating condition has cleared and that the intended next mode is admissible. Separate entry and recovery guards, with persistence checks where appropriate, are proposed to prevent repeated mode switching near a threshold. Their values are not inferred from sampling cadence. Recovery proceeds through Detumble if rate stabilization is needed, otherwise through Sun Pointing and readiness checks before science or relay resumes. Interrupted transfers require integrity and sequence reconciliation before records are retired from persistent storage; the final acknowledgement/retry protocol

remains a PDR interface decision.

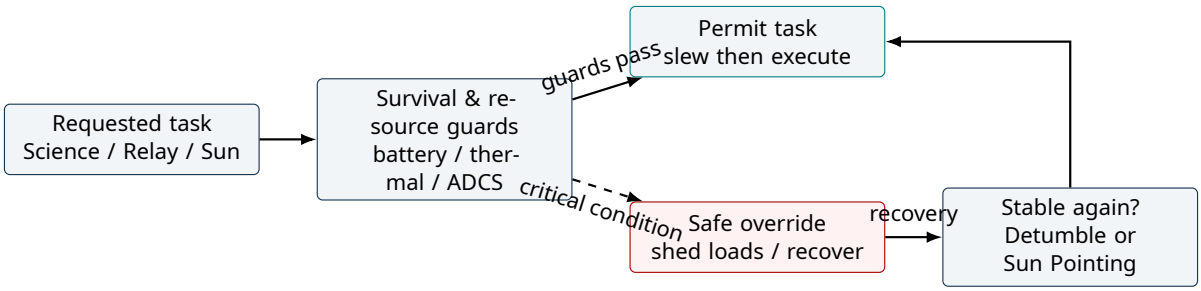


Figure 4: Supervisory decision logic: survival constraints override scheduled science and downlink. Recovery transitions depend on the fault and attitude state; this is a mission-level policy, not an assertion of finalized flight code.

Assumptions and extrapolation limits for A.1.2–A.1.3. The following assumptions make the proposed sequence explicit. They are not additional measured results or approved numerical flight-software settings. The evidence basis is the corrected LEOP/no-LEOP exports and annual run in Appendix A.1.7, together with the current telemetry and communications contracts. Those runs use a historical 2000 epoch, and unresolved telemetry/event timing differences prevent treating this narrative as a reconstruction of every executed transition.

Table 5: Planning assumptions used to extend the operational sequence

ID	Assumed operating interpretation	Basis and required verification
OP-A1	Boot health/time checks and deployment interlocks precede dependent activities.	Necessary interface interpretation; individual checks are not demonstrated by the CSVs. Verify startup software, sensor readiness and deployment inhibits.
OP-A2	Rate/attitude guards release Detumble; Sun Pointing provides the nominal recovery path.	Observed LEOP sequence supports the concept. The 43 min event reference is not a fixed timer or reliability bound; qualify guards and initial-condition coverage.
OP-A3	Scheduling reserves time and energy for slew, settling and task recovery.	Logs contain slew and mode overhead, with differing timing semantics. Reconcile these and establish pointing, reserve and instrument-readiness allocations.
OP-A4	Science and Relay compete for scheduled operation; deferral considers science opportunity and backlog.	Exclusive modeled modes and corrected queue studies motivate the trade. Verify arbitration, outage handling and any future concurrent-operation claim.
OP-A5	Overrides preserve valid interval boundaries and recoverable record state.	Required by nonoverlapping v3 records and integrity objectives; power-loss atomicity and interrupted-transfer recovery are not demonstrated.
OP-A6	Recovery uses fault clearance, persistence and distinct entry/exit guards.	Proposed protection against unstable switching. Set thresholds from component limits and control behavior, then exercise fault injection and repeated recovery.

These assumptions support a preliminary operational concept, not a quantitative guaran-

tee of science yield, safe-mode endurance or communications reliability. Essential pointing, battery-reserve, thermal and readiness limits must be allocated before the sequence can demonstrate feasibility across the required operating cases; hardware tests and detailed implementation verification then close the PDR evidence.

A.1.4 Transition and Guard Matrix

Table 6: Required operational transitions and relevant checks

From	To	Guard / interpretation
Cold & Dark	Detumble	Boot/commissioning delay complete, EPS and OBC alive.
Detumble	Sun Pointing	Body rate falls below accepted rate threshold and attitude solution remains stable.
Sun Pointing	Slewing → Science	SAA opportunity exists, safe battery reserve and payload readiness; +X boresight target feasible.
Sun Pointing	Slewing → Relay	Ground contact above selected mask, queued data or command requirement, sufficient energy; +Y antenna acquisition.
Science	Sun Pointing / Relay	SAA window ends or a higher-priority valid contact is scheduled; use Slewing as needed.
Relay	Sun Pointing / Science	Contact ends, queue complete or higher-priority science becomes available; attitude guard applies.
Sun Pointing	Passive / Eclipse	Eclipse begins; passively retain last attitude rather than repeatedly Sun-searching.
Passive / Eclipse	Sun Pointing	Sun visibility returns and recovery/scheduling permits active pointing.
Any operational mode	Safe	Critical battery, ADCS, thermal or spacecraft health condition supersedes nominal task.
Safe	Detumble / Sun Pointing	Fault clearance and recovery guards satisfied; select Detumble if rates require stabilization.
Slewing	Scheduled target / Safe	Target within pointing/settling tolerance or slew interruption on fault/resource guard.

A.1.5 Timing, Power and Science Opportunity Accounting

A useful SAA crossing is not automatically a valid measurement window. The data-taking interval is the portion of the crossing for which the payload is enabled, pointing has settled, coincidence acquisition is active and no resource/safety override applies. Equivalently,

$$T_{\text{science,usable}} = \int_{t_0}^{t_1} \mathbf{1}_{\text{SAA}}\, \mathbf{1}_{\text{payload}}\, \mathbf{1}_{\text{pointing}}\, \mathbf{1}_{\text{health}}\, dt. \tag{1}$$

The integrated logs distinguish transitions, slews, SAA passages, accepted proton events and ground-station contacts. The current one-year provisional output reports 2,346 SAA-passage records and 820 primary-station contact records. These are *event-log counts for the supplied historical-epoch run*, not unique orbital revolutions or verified 2029 yields. A 2,460-second startup detumble event is observed in that run. These values are to be reconciled with the approved 2029 run before final MCR exit.

A.1.6 Implementation Boundary

The mode architecture, fault-preemption policy, Sun-pointing fallback and science/relay sequencing are MCR-level decisions. Exact battery-voltage thresholds, rate limits, thermal recovery timers, attitude-control gains, antenna acquisition tolerances and software task deadlines belong to the controlled ADCS/OBC configuration and must be verified at PDR. This section intentionally describes the complete concept and its verification boundary without claiming an untested flight implementation.

A.1.7 Observed Startup Evidence

The corrected exports include one 2460 s detumble event in each LEOP scenario, with initial rates $(6, -5.5, 4.5)$ deg/s and accelerations $(0.12, -0.015, 0.08)$ deg/s^2. No-LEOP has no detumble-event schema. This is not a verified deployment distribution. Corrected telemetry spans the complete named horizons, but transition timing and slew occupancy differ from event definitions; mode guards and recovery are not thereby qualified. All three corrected scenarios now participate in the evidence review; the prior exclusion of incomplete day exports is superseded.

A.2 Detailed Scientific Payload: Approved FPDU Concept

The current source of truth is configuration `FPDU-2026-11-04-approved-v2`, referenced by `config/payload/current.json`. It supersedes the earlier same-day input snapshot and the single-block Rev. 2.1 narrative without claiming a formally released hardware revision. The mission requirement remains directional differential proton-flux measurement over 5–100 MeV. No geometry-matched detector response has yet qualified that interval.

Evidence convention. **CONFIRMED DESIGN INPUT** denotes approved construction or mission inputs; **PRELIMINARY ESTIMATE** denotes conditional planning assumptions; **MODEL PREDICTION** denotes a calculation under stated assumptions; **VALIDATED RESULT** is restricted to the tested software identity or arithmetic, not inferred hardware performance; **OPEN DECISION** denotes unresolved design choices. Detector response and hardware performance require validation.

A.2.1 Geometry, Axes and Optical Interfaces

Table 7: Confirmed FPDU mechanical inputs; dimensions exclude packaging

Element	Current specification	Interpretation
Main scintillators	Five, each $14_X \times 6_Y \times 6_Z$ mm	In series along X; 70 mm active length
Front scintillator	One, $2_X \times 6_Y \times 6_Z$ mm	Total nominal active length 72 mm
Main-block SiPMs	One on each $+Y$ and $-Y$ face	Ten main sensors plus one front sensor: eleven total
Main-block reflector	Teflon, 0.11 mm per $\pm Z$ face	Coated Z dimension 6.22 mm; X and Y faces not coated by default
Optical coupling	EJ-550 grease	Thickness, coupling area and sensor models open
Entrance shield	250 μm = 0.25 mm Al	Front entrance only; transport not validated
Silicon / rear detector	Silicon removed; rear optional	Optional rear scintillator not selected or included in dimensions

The body-frame $+X$ direction points outward along the RAM/payload boresight. Particles enter the $+X$-facing entrance and propagate inward toward $-X$. All six scintillators are intended to share one material; its selection is open. Material volume is 504 mm^3 per main block, 2520 mm^3 for five, 72 mm^3 for the front, and 2592 mm^3 = 2.592 cm^3 in total. These exclude gaps, coating, grease, shielding, sensors, supports and any rear element. Neither packaged length nor payload mass follows from active dimensions alone.

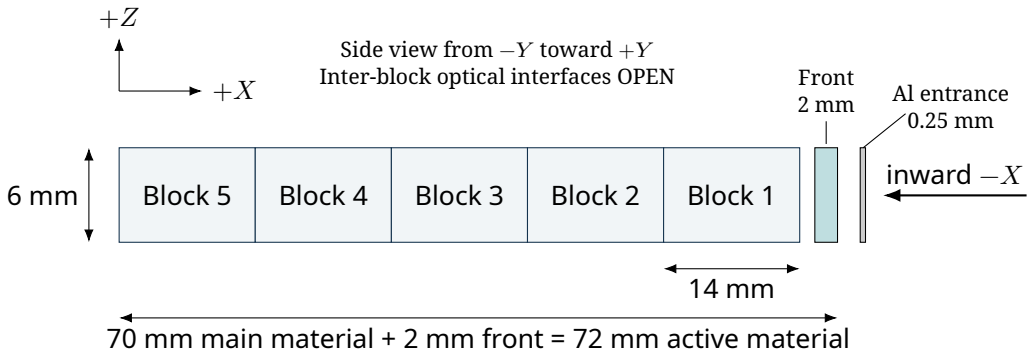


Figure 5: Longitudinal concept, not a packaged CAD drawing. Shown gaps and shield thickness are schematic. Dimension sums count active material only. An optional rear detector would lie beyond Block 5 and is not included.

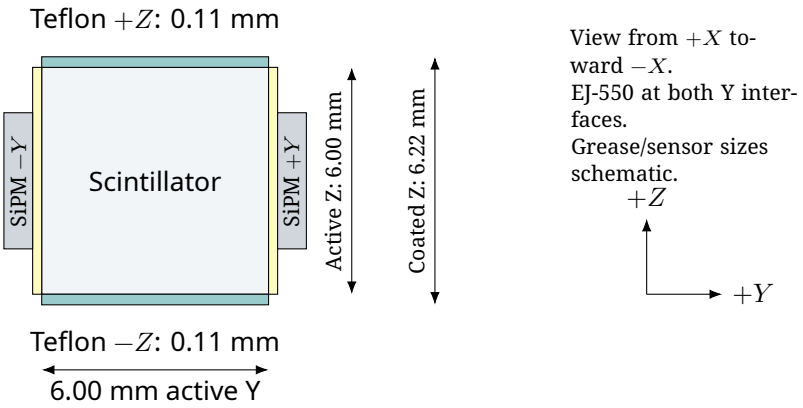


Figure 6: Main-block cross-section. Optical isolation between successive X faces is undecided. The front scintillator's coating and SiPM attachment face remain open; this cross-section does not prescribe them.

OPEN DECISION: optical isolation. Independently isolated blocks could localize deposition but require defined separators and per-block calibration. Partial longitudinal coupling introduces light sharing and cross-talk that must be measured before using multiplicity or position observables. Common coupling, if physically feasible, would require a calibrated light-sharing model and may reduce deposition localization. None is selected. Surface finish, inter-block material, EJ-550 thickness, photon collection, SiPM saturation and temperature behavior remain unqualified. No controlled current FPDU CAD/ICD was recovered.

A.2.2 Signal Chain and Timing Trade

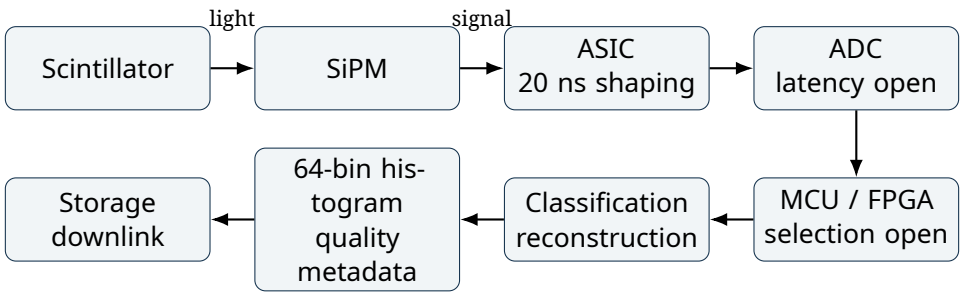


Figure 7: Candidate signal and science-data flow. EJ-550 provides the optical interface. Eleven SiPMs do not specify eleven ADCs. Fast coincidence may occur in analog/ASIC logic while downstream processing is buffered; this architecture remains to be selected.

Table 8: Separate timing references and missing qualification inputs

Quantity	Planning input	Interpretation
ASIC shaping time	20 ns	Pulse-shaping parameter, not coincidence or dead time
Coincidence target	200 ns	Nominal acceptance-window reference
Initial timing allowance	≥ 250 ns	Margin-inclusive window reference
Preferred allowance	~ 300 ns	Comfortable window reference; accidental-coincidence trade required
Post-shaping delay	20 ns stated	Meaning unresolved; not assigned as ADC conversion latency
ADC conversion / processing	Unknown / unknown	Datasheets, pipeline and worst-case execution measurements required
Qualified detector dead time	Unknown	Shared/per-channel topology and loss model unresolved
Timestamp precision / jitter	Unknown	Channel skew and discriminator/capture measurements required
Maximum trigger throughput	Unknown	All-particle acceptance, burst and queue tests required

The 200/250/300 ns values are coincidence-window planning references, as clarified by the team. They are not processor execution deadlines. Widening a window changes accidental coincidences; it does not prove engineering margin. The two stated 20 ns values must not simply be summed into an end-to-end latency. No dead time is assigned from any of these values.

MCU is preferred only if the complete acquisition architecture demonstrates adequate timestamp precision, coincidence handling, ADC bandwidth, buffering, processing throughput and power. Compare ASIC/discriminator coincidence with MCU processing, timer/capture-assisted MCU, FPGA, and hybrid FPGA/MCU implementations. For a stated mean arrival rate λ and service time s, $\lambda s < 1$ is a queue-stability prerequisite, not proof of low event loss. Finite buffers and burst arrivals need explicit tests. Coincidence, ADC acquisition, service time, event loss and detector livetime remain separate quantities.

A.2.3 Radiation Inputs and Conditional Rate Estimates

The supplied AP9/AE9 maps are integral omnidirectional fluxes at MJD 61041 (1 January 2026), altitude 600 km. They are team-approved preliminary references for 2029 mission planning, not regenerated 2029 environment outputs. The rate conversion assumes isotropy and $G = 1$ cm^2 sr with unit efficiency: $R = F_{\text{omni}} G/(4\pi)$. It uses log–log interpolation at 5.5 MeV for protons and 0.35 MeV for electrons. The script contains no aluminium transport, actual telescope acceptance, electron rejection, optical response or dead-time model.

Table 9: Reproduced MODEL PREDICTIONS: idealized incident-rate proxy maxima (Hz)

Map	Proton	Electron	Interpretation
Mean	358.920	67 779.655	Nominal preliminary environment
99th percentile	975.659	503 419.743	Conservative study scenario, not a hard upper bound

The species maxima need not occur at the same cell. Maximum same-cell summed proxies are

68 055.635 Hz (mean) and 504 245.064 Hz (99th-percentile maps); the latter is not a joint 99th-percentile trigger bound. Values are independently reproduced from `data/raw/payload_count_rate_2026/` in the count-rate validation report. Flux-times-area proxy columns must not be confused with the columns divided by 4π.

PRELIMINARY ESTIMATE, superseded rounded reference: approximately 400 Hz assumed protons only, 250 μm (0.25 mm) aluminium shielding, geometric factor 1 cm^2 sr and 100% detection efficiency. Its original generating calculation remains unidentified. The supplied per-cell rates take priority; they do not implement the stated aluminium shielding. No constant rate or electronics ceiling is imposed.

The proton source grid ends at an integral threshold of 20 MeV; its above-threshold quantities are not truncated differential bins. It cannot supply a calibrated 5–100 MeV spectrum or an integral subtraction at 100 MeV. Electrons may consume front-end resources before downstream rejection. A credible maximum actual trigger rate requires geometry, transport, thresholds, mixed species and burst evidence and is not established here.

A.2.4 Reconstruction, Histogram and Telemetry Contract

The front scintillator is intended to respond in the proposed 5.5–15 MeV incident range and provide deposition, path-length-corrected dE/dx, and classification/coincidence observables. These functions are not calibrated incident-energy measurements. Response-matrix lookup/unfolding and a trained reconstruction model remain candidates. Evaluation must use detector observables, keeping truth energy only as the reference target. The old five-layer stopping response and silicon-based coincidence do not apply.

Table 10: Science Record v3: 292-byte uncompressed reference, excluding radio framing

Field	Bytes	Reference encoding
Entry / exit timestamps	16	Two uint64 microsecond POSIX UTC values since 1970-01-01; half-open interval
Latitude / longitude	8	Two signed int32, degrees $\times 10^7$; midpoint WGS84 geodetic position
Altitude	4	Signed int32 millimetres above WGS84 ellipsoid
64 histogram bins	256	uint32 integer counts, provisional 5–100 MeV reporting edges
Sequence	4	uint32; reboot/wrap epoch in session manifest
Combined status	2	uint16 mode and six subsystem-health pairs
CRC16	2	CRC-16/IBM-3740 over preceding 290 bytes
Total	292	Little endian, no padding; no radio overhead included

The reference definition is `config/science/science_record_v3.json`. Version, bin-edge ID `FPDU-log64-5-100-v1`, calibration/response/configuration identity and time correlation are mandatory linked session metadata, not hidden extra bytes in the record. Actual livetime, losses, background and quality details require linked interval metadata whose size remains unbudgeted. The 64 bins are not 64 independently calibrated channels. Fractional modeled expectations are not valid uint32 measured counts.

CRC polynomial is 0x1021, initial value 0xFFFF, nonreflected, final XOR zero; CRC bytes are little endian. The reference check for “123456789” is 0x29B1. UTC leap seconds are not representable in POSIX time; split/flag affected acquisitions and retain clock correlation. Encoding resolution is not clock or navigation accuracy. No final radio packet size or throughput qualification is implied.

Table 11: Combined 16-bit mode and health codebook

Bits	Meaning
0–3	0 Cold/Dark; 1 Detumble; 2 Sun Pointing; 3 Detection; 4 Relay; 5 Slewing; 6 Safe; 7–14 reserved; 15 unknown
4–5 / 6–7 / 8–9	OBC / EPS / ADCS health
10–11 / 12–13 / 14–15	COMMS / payload / other-thermal health
Each health pair	00 nominal; 01 degraded; 10 fault; 11 unknown

These new serialization codes map the actual FSM names, not pre-existing numeric telemetry values. Unknown health must not default to nominal. Detailed diagnostics are separate; this marker cannot replace calibrated exposure or background information.

Reading validity and accumulation. Prefer three readings per eligible geographic cell visit, with two or one when available timing permits; no valid exposure means no record. No arbitrary minimum duration or count threshold is imposed. Each reading has its own positive, nonoverlapping interval, geographic attribution and valid acquisition state. Split at cell/mode boundaries and shutdown; do not bridge off periods. Valid zero-count records are retained. Realizable boundaries depend on timer resolution, readout latency and uninterrupted counter handoff, all still unqualified. A short interval can be a valid record without achieving a mission precision target.

For an ideal zero-background Poisson diagnostic, accumulated fractional counting uncertainty is $1/\sqrt{\sum_i N_i}$, not an average of individual fractional errors. Background, livetime and reconstruction uncertainty remain necessary. Zero-count exposures contribute to inference and must not be discarded. Splitting a visit preserves total exposure and counts and does not create independent orbital passages. Approved MMSC/MSC/EMSC requirements are unchanged. The offline reference partitions closed acquisition segments; the flight scheduler and event-buffer/counter-bank design require PDR verification.

For directional differential flux Φ_s in cm^{-2} s^{-1} sr^{-1} MeV^{-1} and effective area $A_{j,s}$ in cm^2, the response contract is

$$N_j = \int \ell(t) \sum_s \int\int A_{j,s}(E,\Omega,t)\Phi_s(E,\Omega,t)\, dE\, d\Omega\, dt. \tag{2}$$

A geometric factor already includes the angular integral. Shared-trigger livetime, energy migration, background subtraction and calibration uncertainty require a qualified model. Geometric coverage or synthetic counting precision alone cannot establish MMSC, MSC or EMSC.

A.2.5 Geometric-Factor Requirement from Cell Counts

MODEL PREDICTION; arithmetic independently reproduced. The supplied geometric-factor study asks how much ideal proton acceptance is needed to reach a counting-only precision target in each cell. It uses 7940 unique 0.5° polygon-overlap cells, integral-spectrum differences and supplied cell dwell times of 98–294 s. The dwell file matches an archive labelled “18-Month Simulation”, but the executed configuration and duration are missing. Nearby source defaults to three days and cannot explain the accumulated table. No mission-duration scaling is applied. The radiation table is AP9-labelled; its actual mean/percentile setting, epoch and 65-threshold source run remain unverified. It is distinct from the verified 2026 AP9/AE9 maps above.

Under isotropy, unit efficiency and a scalar geometric factor,

$$R_{u,c,b} = \frac{F_c(>E_{b,\mathrm{lo}}) - F_c(>E_{b,\mathrm{hi}})}{4\pi}, \qquad N_{c,b} = R_{u,c,b} G \epsilon T_c, \qquad G_{\mathrm{req},c,b} = \frac{1}{\sigma^2 R_{u,c,b} \epsilon T_c}. \tag{3}$$

Here F is omnidirectional integral flux in cm^{-2} s^{-1}, R_u is rate per unit G in s^{-1}/(cm^2 sr), T is supplied dwell in seconds, and G is in cm^2 sr. No extra energy width or angular integral is applied. The expected-count targets for $\sigma = 10, 15, 25, 50\%$ are 100, 44.44, 16 and 4; these are not total reconstructed-flux uncertainties or guarantees on realized counts.

Table 12: Reproduced required G_{95} for each separately selected cell population; $\epsilon = 1$, cutoff 5.4 MeV, supplied dwell. All G values are cm^2 sr.

Band (MeV)	Cells	10%	15%	25%	50%
5.4–10	7596	0.364535	0.162016	0.058326	0.014581
10–20	7177	0.413303	0.183690	0.066129	0.016532
20–40	7052	0.500042	0.222241	0.080007	0.020002
40–100	7144	0.360776	0.160345	0.057724	0.014431

Each band selects cells with rate at least 1% of its own peak and takes the linearly interpolated 95th percentile of their required G. This resembles one ingredient of EM-M-02, but does not establish its mission-peak eligibility definition or joint channel coverage. At the hypothetical $G = 0.3$ cm^2 sr, the 10% counting-target coverages are 93.55, 92.25, 89.66 and 93.25% of these respective populations. All four meet the 25% expected-count target in their selected cells. These are not mission-success percentages.

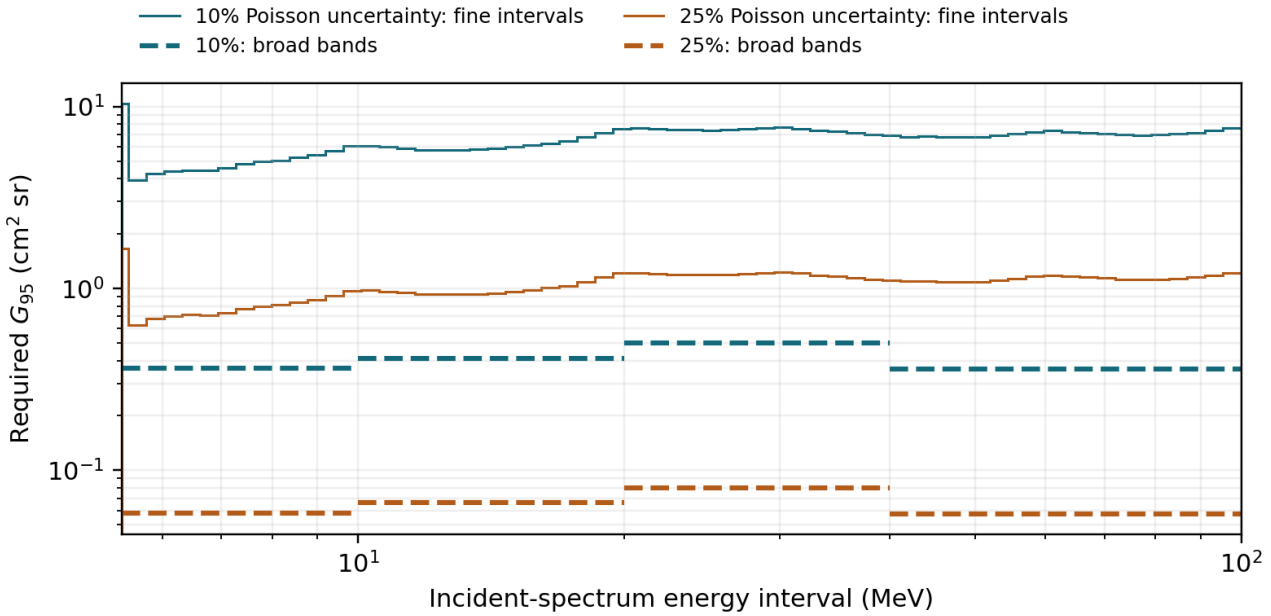


Figure 8: Independent reproduction of the required-G study. Fine intervals require more acceptance for the same counting precision than broad bands; each interval has its own eligible-cell population. This is a count requirement, not an achievable or measured FPDU response.

The input has 65 thresholds over 5–100 MeV, giving 64 intervals. The 5.4 MeV cutoff removes the first and truncates the second, leaving 63 intervals. Their 10% G_{95} values span 3.9284–10.3301 cm^2 sr, compared with 0.3608–0.5000 for the broad bands. Neither is evidence for 64 resolved channels. The provisional 64-bin telemetry retains its separate 5–100 MeV definition; merging reporting bins does not establish the required energy resolution.

Required versus achievable acceptance. An ideal unrestricted one-sided 6 mm square aperture has $G = \pi A = 1.13097$ cm^2 sr. Two equal aligned planes separated by 2 mm, 4.8 mm or 72 mm give approximately 0.61907, 0.29743 or 0.0024885 cm^2 sr, respectively. These are hypothetical straight-track acceptance conditions: no collimator, material or all-block coincidence rule is selected. Crossing both ends of a gap-free aligned 72 mm assembly implies crossing every block, but early-stopping or side-exiting particles need different acceptance logic. The supplied optional model assigns on-axis stopping depths using illustrative EJ-200 and a simplified CSDA approximation; it omits angular transport, scattering, straggling and qualified triggers. It cannot establish the actual energy-dependent effective G. Its coarse quadrature also overestimates the artificial 0.2 mm separation case by about 3.6%.

Cutoff, exposure and PDR conditions. The default 5.4 MeV cutoff is a hard study assumption, not aluminium transport. The optional source approximation instead derives about 5.95 MeV for 250 μm Al; neither value is a validated transmission threshold. Changing only the counting cutoff to 5.5 MeV increases first-band $G_{95}(10\%)$ to 0.380454 cm^2 sr (4.37%), with 7588 selected cells. The committed 5–100 MeV requirement is unchanged. Required G scales inversely with effective efficiency and verified exposure; no duration extrapolation is justified by the current manifest gap. Background subtraction, livetime, reconstruction covariance, independent passages, temporal maps and stored/downlinked records must be validated before tier acceptance. Source hashes, reproduced tables and tests are indexed in `docs/audit/GEOMETRIC_FACTOR_NUMERICAL_VALIDATION.md`.

A.2.6 Conditions for PDR Closure

Table 13: Remaining verification conditions

Interface	Required closure evidence
Mechanical / optical	Controlled CAD/ICD, material, front treatment, gaps, optical isolation, grease and SiPM placement; measured cross-talk and collection maps
Transport / acceptance	Current six-scintillator geometry, entrance Al, proton 5–100 MeV and electron/species/angular scans; validate effective geometric factor
Electronics	Selected channels/components, post-shaping interpretation, timing/jitter, queue/burst tests, measured dead time and loss counters
Reconstruction	Independent calibration and test data, stopping/penetrating response, mixed-species confusion matrix and injected-spectrum closure
Mission coupling	Science-enabled exposure, qualified channels/background, independent-observation contract and stored/downlinked records before tier acceptance

A.3 Simulation Configuration and Provenance

This appendix records the simulation code used to support the ANTARA mission concept. The simulation scripts are organised by study under `src/simulations/`. The integrated science simulation uses a common configuration defined in `src/simulations/helper/antara_models.py`; the standalone orbit, power, pointing, detumble, CONOPS and thermal studies have their own study-specific configurations. Their results must not be treated as outputs of one identical spacecraft model unless the input parameters have been checked.

A.3.1 Simulation Inventory

Table 14: Simulation source files and principal applications.

Study	Python entrypoint(s)	Intended evidence
Integrated science and DITL	`ditl/ANTARA_Science_Sim.py`; `ditl/scenarios.py`; `ditl/compare_scenarios.py`	Mission modes, power, attitude, SAA science opportunities and ground contact
Orbital decay	`orbit/ANTARA_drag_finalconfig.py`; `orbit/ANTARA_drag_study.py`	Orbital decay and lifetime studies
Power	`power/ANTARA_MCR.py`; `power/ANTARA_avg.py`; `power/ANTARA_FINAL_POWERGEN.py`; `power/TradeCase.py`	Power-generation and battery trade studies
Pointing	`pointing/ANTARA_pointing_budget.py`	Ground-station pointing geometry
Detumble	`detumble/leop_detumble_mc.py`; `detumble/analyze_detumble_mc.py`	Initial-rate reduction and Monte Carlo statistics
CONOPS	`conops/ANTARA_conops.py`	SAA and station encounter analysis
Thermal	`thermal/ThermalSimulator.py`	Lumped-parameter temperature studies
Shared support	`helper/antara_models.py`; `helper/fsm_spec.py`; `helper/telemetry_metrics.py`; other `helper/` modules	Configuration, attitude models, FSM rules and telemetry processing
Historical	`legacy/ANTARA_Master.py`	Superseded simulation; not a current baseline

A.3.2 Integrated Simulation Baseline

The values in Table 15 are the defaults in `MissionConfig` supplied with the reviewed source archive. These are simulation inputs, not final hardware qualifications or independent performance measurements. A command-line override or a run-specific configuration can change them.

Table 15: Selected integrated simulation configuration defaults.

Parameter	Code-defined value	Unit
Initial orbital altitude	600	km
Inclination	97.6	degrees
Eccentricity	0.001	—
RAAN	15	degrees
Initial spacecraft mass	3.0	kg
Drag coefficient	2.2	—

Parameter	Code-defined value	Unit
SRP coefficient	1.8	—
Simulation start date	1 March 2000	UTC calendar date
Nominal fine time step	2	s
Coarse time step	20	s
Battery planning capacity	25	Wh
Initial stored battery energy	18	Wh
EPS charge efficiency	0.88	—
Solar temperature derating	0.88	—
Maximum solar generation parameter	22	W
OBC load parameter	0.8	W
EPS load parameter	1.2	W
Payload load parameter	10	W
Radio transmit load parameter	12	W
ADCS idle and slew loads	0.8 / 4.5	W
Radio downlink rate	0.0192	Mbit/s
Primary ground-station location	15° N, 74° E	coordinates
Ground-station altitude	70	m
Ground-station elevation mask	15	degrees
Random seed	42	—

The integrated model also defines power thresholds for entering and exiting low-power conditions, attitude-pointing limits, detection and relay minimum durations, a maximum wheel-saturation hold, and fault/recovery debounce settings. The exact configuration used for a reported run must be retained with that run, rather than inferred only from current defaults.

A.3.3 Integrated DITL Scenarios

The three standard presets are defined in `ditl/scenarios.py` and shown in Table 16. The two 24-hour cases separate the startup sequence from nominal operations; the one-year preset includes startup.

Table 16: Integrated Science simulator presets.

Preset	Duration	Starting condition
`leop_24h`	24 h	Includes cold-start and detumble sequencing
`no_leop_24h`	24 h	Skips LEOP and starts in Sun Pointing
`leop_1yr`	8,760 h	Includes LEOP; uses multirate integration with a configured 20 s coarse step

Configuration precedence in the source is explicit command-line options, followed by scenario preset, followed by the global defaults. In particular, the RAAN and battery capacity are configuration fields and are not exposed as direct `--raan-deg` or `--battery-wh` command-line options in the documented Science entrypoint.

A.3.4 Numerical and Physics Models

The integrated simulator uses Orekit numerical orbit propagation and includes gravitational perturbations, an atmospheric drag model and solar radiation pressure. The source im-

ports the Holmes–Featherstone gravity-field model and the DTM2000 and NRLMSISE-00 atmospheric models. The model and options active in a particular run must be established from the saved configuration and executed code path.

The integrated model uses scalar-first quaternions, $[w, x, y, z]$, for body-to-inertial attitude transformations. The spacecraft body axes are defined with $+X$ along the payload boresight, $+Y$ toward the relay antenna and $+Z$ along the fixed solar-panel face. The simulation includes finite slews and pointing-settlement checks; an attitude command does not produce an instantaneous orientation change.

The magnetic-field implementation supports the configured dipole option and an IGRF backend through `ppigrf`. If the optional IGRF package is unavailable, the documented IGRF path warns and falls back to a centred dipole. This fallback is not equivalent to an IGRF-based result and must be recorded when used.

The proton environment model applies SAA spatial variation, magnetic pitch-angle dependence, east–west asymmetry, storm scaling and detector response to a configured baseline spectrum. Although a legacy interface and some telemetry keys use the name ap9, the present source does *not* execute a live AP9 calculation. Consequently, simulator flux values must not be represented as independently generated AP9 predictions or validated in-orbit measurements. The simulator's detector counts are modeled expected counts, rather than a calibrated hardware measurement campaign.

A.3.5 Time Accounting and Data Outputs

The Science simulator writes scenario-tagged CSV files under `results/science/`, including telemetry, SAA passes, ground-station contacts, mode transitions, slew events, detumble events, rejected transitions and proton passes. Figures are written under `figures/science/` when plot generation is enabled.

Newer source telemetry distinguishes the row observation time, `time_sec`, from the integrated interval end, `state_time_sec`. `interval_duration_sec` records the elapsed integration interval. Those interval columns are absent from the corrected exports. This audit uses successive recorded timestamps, assigns no duration after the endpoint, and retains timing-semantic uncertainty. In the newer source, the final terminal snapshot has zero duration. Time-averaged quantities and mode occupancy must therefore be weighted by the interval durations rather than the number of CSV rows. Plot sampling is for display only; extrema and scientific statistics should be calculated from complete telemetry and validated summary outputs.

The simulator supports trusted, versioned checkpoints and resumable execution. Checkpoints and their linked science-event archives must be retained together. A resume record does not by itself establish that the entire intended scenario completed correctly.

A.3.6 Supporting Study Differences

The standalone scripts should be treated as supporting studies rather than interchangeable implementations of the integrated Science scenario. In particular:

- The standalone pointing-budget study uses its own nadir-pointing and antenna-boresight assumptions; these must be compared with the integrated $+Y$ relay antenna geometry before a result is used as the spacecraft relay pointing budget.
- Standalone solar-power and battery studies contain their own panel and operating assumptions, which must be reconciled with the integrated Science configuration.

- Detumble Monte Carlo trials are useful for testing the controller over initial conditions, but do not automatically demonstrate integrated EPS, payload and ground-station operation.
- The thermal model is a separate lumped-parameter study, not a temperature field directly solved within every Science simulator step.
- The historical `ANTARA_Master.py` is superseded and should not be used as the current simulation baseline.

A.3.7 Verification Status and Known Limits

The supplied repository's `docs/CURRENT_STATUS.md` records a 414-test software regression run during Phase 3B and matching one-hour Science baseline outputs. These are repository-reported verification results; they have not been independently reproduced in preparing this appendix. The same record states that a 60-hour annual-preset run does not establish completion or validation of the full one-year scenario. It also identifies the absence of live AP9 calculation and the superseded status of the Master implementation.

For each figure or numerical conclusion adopted in the MCR, retain the following alongside the output: source revision or commit, scenario name, complete run configuration, RNG seed where applicable, simulator start date, run duration and completion status, numerical step controls, output path, and any modifications to the baseline model. Without these details, a plotted quantity cannot be reliably associated with a particular spacecraft design configuration.

A.3.8 Simulation Evidence Record

The following records distinguish approved planning inputs from verified generating epochs. Full claims and hashes are indexed in the audit reconciliation and count-rate ingestion reports.

Table 17: Current evidence identities and limits

Evidence	Identity	Acceptance limit
Mission baseline	`ANTARA-2029-v1`; 2029-01-01	Approved configuration, not completed run
Radiation rate maps	MJD 61041; 600 km; AP9/AE9 mean and 99th percentile	Reproduced arithmetic; 2026 source accepted for preliminary 2029 planning
RAAN study	Seven cases, two runs each; supplied CSVs	Team-accepted 2029 reference; generating epoch not independently verified
Historical DITL	March 2000 source run	Historical operational example, not current payload qualification
Corrected Science exports	Two 24 h and one 365-day timestamp spans; March 2000	Runtime source/configuration and mode semantics unresolved
Required-G study	7940 cells; 65 integral thresholds; supplied dwell	Tables reproduced; source epoch, percentile and exposure duration unverified
Current FPDU	Approved-v2 configuration	Mechanical inputs and logical interfaces only; no qualified response

Payload and software reconciliation. The current configuration is not consumed by the legacy DITL/Geant4 response. Eleven SiPMs, removed silicon, open optics and unknown dead time require geometry-matched validation. Coincidence references are 200/250/300 ns; shaping is 20 ns. Historical outputs remain unchanged. Bounded tests validate accounting only. Tested source fixes are delivered as an unapplied patch because the authoritative simulator is outside the writable workspace.

A.3.9 New Operational, Thermal and Detumble Evidence

Corrected Science authority is controlled by `evidence/science-ditl/run_registry.json`; prior Science manifests are superseded. Independent study manifests remain under `evidence/science-ditl/manifests/`, `evidence/thermal/manifests/` and `evidence/detumble/manifests/`. Current external source hashes are candidate references only; no generating commit or runtime manifest was supplied. Command-line duration overrides cannot be recovered from preset names.

Table 18: Observed evidence completeness; no expensive simulation rerun

Evidence	Observed span	Acceptance boundary
Corrected LEOP / No-LEOP	7696 / 6553 rows; 0–86400 s	Complete timestamp spans; runtime configuration unknown
Corrected annual telemetry	1434872 rows; 0–31536000 s	Complete365-day timestamp span; interval semantics conditional
Corrected annual events	2000-03-01 epoch	Contact/data totals reconcile; mode/slew definitions require closure
Thermal	43 201 labels; 0–24 h; 2 s	One history; no separate HOT/COLD cases; initial-state timing unresolved
Detumble	50 000 summary rows	Historical inputs incompatible with current source; censor times/seed absent

Current candidate orbit and thermal integrators are DP853 and RK4 respectively, but defaults cannot establish the producing solver settings. The thermal heat schedule is independent of DITL. Passing local accounting tests establishes parsing, interval weighting and summary arithmetic only; it does not validate current hardware response or a common 2029 run.

Science-record and communications revision. Science Record v3 and the full 19.2 kbps capacity analysis are new MCR reference interfaces, not outputs of the historical simulator. Source hashes and exact per-contact/tier calculations are under `data/derived/ditl_correction_2026-11-09/`; the generating scripts read originals without changing them. The grid-study horizon remains unknown, annual contacts retain their 2000 epoch, and corrected telemetry spans365days; the old partial artifact is superseded. The 292-byte reference supersedes older unencoded record-size assumptions without modifying detector physics, scientific thresholds or original transfers.

A.4 Thermal and Structural Supporting Analysis

This appendix describes the thermal model implemented in `src/simulations/thermal/ThermalSimulator.py` and the structural information available in the MCR. The thermal results are model predictions; no spacecraft-level thermal-vacuum or mechanical qualification is claimed.

A.4.1 Thermal Model and Spacecraft Geometry

The simulator represents the spacecraft with seven lumped-capacitance nodes: the six external faces and an internal bus. The geometric model uses a $10\times10\times34$ cm 3U body with the long axis along X. Each end face has area $0.010\ \mathrm{m}^2$ and each long face has area $0.034\ \mathrm{m}^2$. Each node carries mass, specific heat, solar absorptivity, infrared emissivity and thermal conductance to the internal bus. Neighboring external faces are also coupled through a specified conductance. These values form a preliminary thermal representation, not measured properties of the final assembly.

Table 19: Selected thermal simulation inputs (`ThermalMissionConfig` defaults).

Input	Code setting
Initial orbit altitude / inclination	600 km / 97.6°
RAAN and eccentricity	15° / 0.001
Spacecraft model mass	3.0 kg
Simulation duration / nominal time step	24 h / 2 s
Thermal substeps per simulation step	10
Initial temperature	293.15 K
Internal bus lumped mass / specific heat	1.8 kg / 800 J kg^{-1} K^{-1}
Face-to-face thermal conductance	0.05 W K^{-1}
Solar constant / deep-space temperature	1361 W m^{-2} / 2.7 K
Applied temperature margin parameter	10°C

A.4.2 Thermal Balance and Numerical Method

For each node i, the model integrates a lumped heat balance of the form

$$C_i\frac{\mathrm{d}T_i}{\mathrm{d}t} = Q_{\mathrm{solar},i}+Q_{\mathrm{albedo},i}+Q_{\mathrm{EarthIR},i}+Q_{\mathrm{internal},i}-Q_{\mathrm{radiated},i}+\sum_j G_{ij}(T_j-T_i), \qquad C_i = m_i c_i. \tag{4}$$

The code calculates incident solar radiation, Earth albedo and infrared heating, thermal emission, internal dissipation and conduction. Solar-cell electrical conversion is included in the face heat balance. The orbit is propagated with Orekit numerical integration; the thermal equations use fourth-order Runge–Kutta integration with configurable thermal substeps. Earth albedo and infrared terms may vary with latitude in the implemented model.

A.4.3 Nominal, Hot and Cold Cases

The source defines `NOMINAL`, `HOT` and `COLD` bounding-case configurations. Nominal retains the base configuration. The HOT case places the model near perihelion, increases the end-of-life optical assumption and adds 0.5 W of bus dissipation. The COLD case uses an aphelion

epoch, a lower optical assumption and 0.5 W less bus dissipation; relay and detection loads are disabled for that case. These are deliberately selected model cases, not measured flight extremes. The simulation can also use LVLH or prescribed tumbling attitude. The default internal heat schedule is independent of the Science DITL flight software: it includes a 1.5 W baseline bus, 0.4 W charging term, 0.8 W eclipse-heater term, a 3.0 W relay-pass increment and a 1.2 W detection-window increment with configured durations. Consequently, thermal case temperatures should not be described as arising from the exact commanded state history of `ANTARA_Science_Sim.py`.

A.4.4 Temperature Outputs and Required Comparison

The supplied history contains 43 201 unique time labels from 0 to 24 h. Kelvin/Celsius pairs and all summary extrema/timestamps reproduce independently. Table 20 reports the recorded envelope. Time-weighted means use trapezoidal integration over labels, without adding an interval beyond 24 h. Nonuniform first logged temperatures differ from the candidate 293.15 K initialization; the exact generating source and possible state-label offset require reconciliation.

Table 20: Single-history node statistics; degrees Celsius. No component compliance margin is established.

Node	Minimum	Maximum	Time-weighted mean
+X	-20.775	19.914	-8.907
-X	-17.220	26.911	-5.480
+Y	-15.865	19.951	-9.507
-Y	-25.958	31.239	-7.625
+Z	-22.401	29.601	0.718
-Z	-23.301	19.919	-16.552
Internal	-14.061	21.159	-6.048

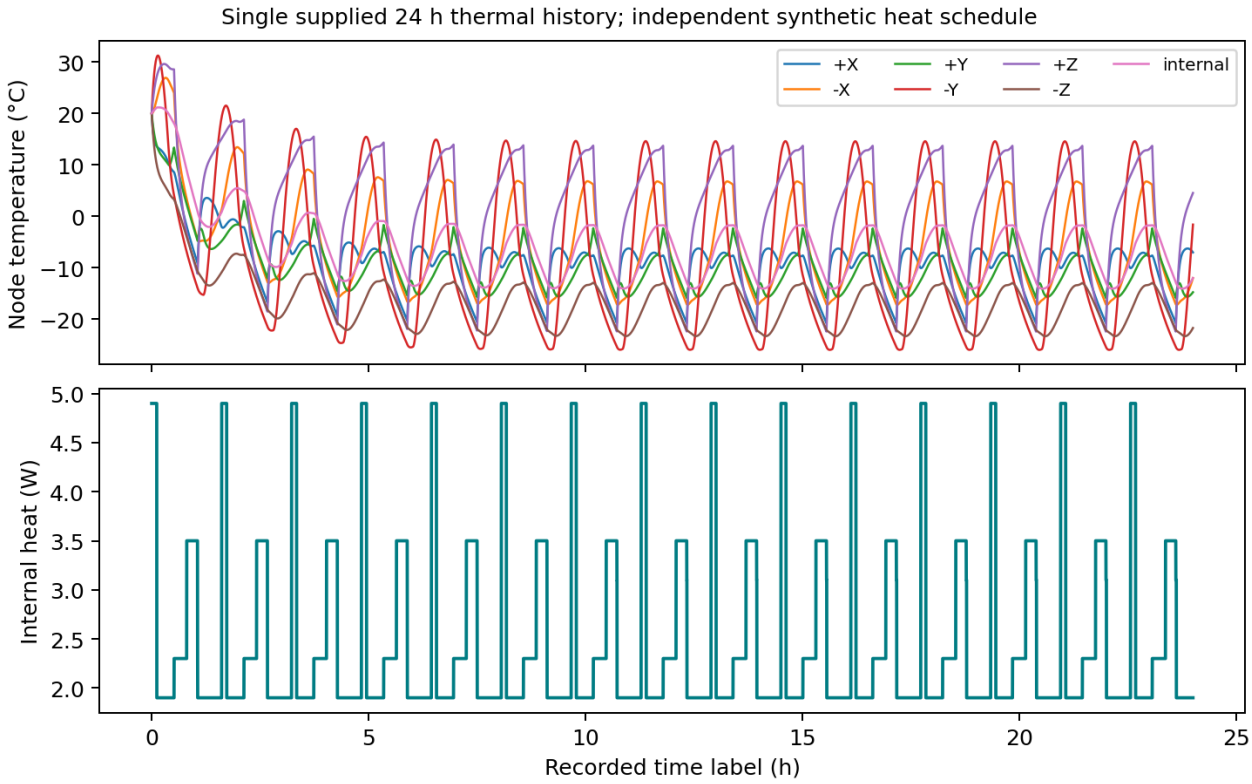


Figure 9: Supplied single thermal history and independent synthetic heat schedule, 1.9–4.9 W. Model predictions; actual case epoch, component mapping and DITL coupling are unverified.

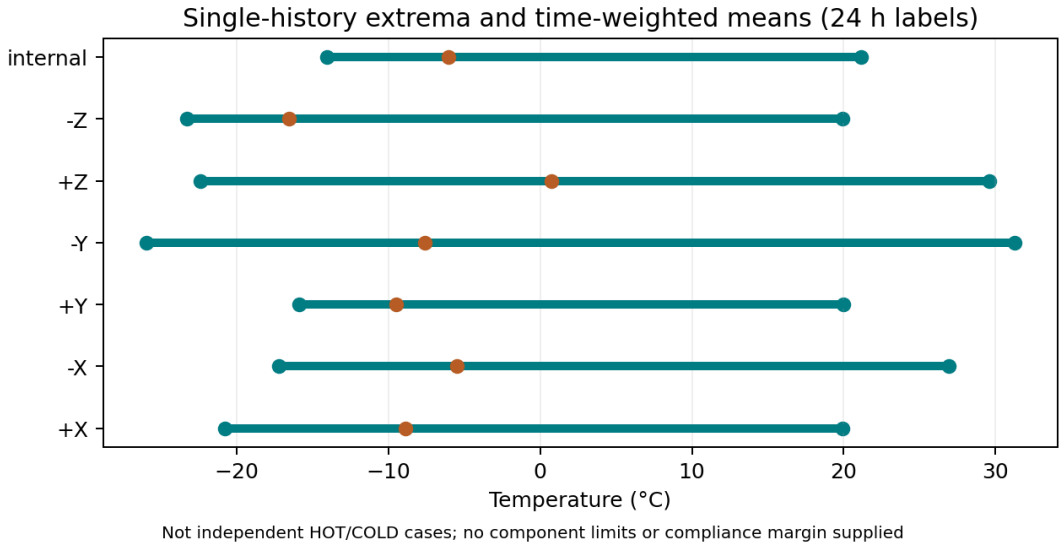


Figure 10: Reproduced minima, maxima and weighted means from the same history. The summary's hot/cold names do not establish separate bounding-case runs.

The largest simultaneous external-face spread is 31.704°C; this is a lumped-node temperature difference, not structural stress. The summary's ± 10°C columns are arithmetic allowances. No selected-component temperature limits are supplied, so time outside qualified limits and compliance margins are unknown. Separate bounding-case manifests, timestep/energy-balance convergence, node-component mapping, heater control, coupled dissipated-power histories and thermal-vacuum correlation remain required.

A.4.5 Structural Configuration and Verification

The current MCR specifies a 3U spacecraft envelope of approximately $100 \times 100 \times 340.5$ mm, a one-fixed/two-deployable-panel configuration and a preliminary component mass table. Structural verification will require the selected structure and deployer interface specification, consolidated mass and inertia properties, solar-panel deployment design and the applicable launch mechanical environment. The provided simulation archive does not establish a completed structural finite-element, modal, vibration or shock qualification analysis. These items therefore remain planned verification work, rather than numerical results to be inserted into the report.

A.5 Detailed Engineering Budgets and Bill of Materials

This appendix records the calculation method and the presently available design inputs for the spacecraft budgets. The existing itemised mass and cost tables are retained in Appendix E. The simulation constants below must not be substituted for component datasheet values without identification.

A.5.1 Component Baseline and Mass Budget

The original draft lists the integrated ADCS, OBC, one body-mounted panel, deployable solar panels, EPS and battery, CubeSat structure, radio, directional and omnidirectional antennas, GNSS receiver and antenna, deployment interface and custom payload. Payload and some mechanism masses and several prices remain unfilled. Because the original component list is not yet a reconciled assembly-level budget, a total wet mass and center of gravity cannot be certified from it.

$$m_{\text{total}} = \sum_i n_i m_i + m_{\text{harness}} + m_{\text{mechanisms}} + m_{\text{contingency}}. \tag{5}$$

For each entry, record quantity, unit mass, selected or proposed part number, data source and whether the quantity is measured, quoted or estimated. The 4.0 kg mass used in the annual RAAN campaign and orbital-decay standalone study differs from the 3.0 kg integrated DITL and thermal defaults. Neither should be silently reported as an as-built mass.

A.5.2 Solar Generation and EPS Budget

The standalone annual RAAN study `power/ANTARA_MCR.py` uses one fixed and two deployed solar panels with fixed area 0.030 m^2, deployed area 0.030 m^2 per wing, cell efficiency 0.28 and a system efficiency factor 0.513. It assumes 8.0 W constant load and a 25 Wh battery. The separate 24-hour `ANTARA_FINAL_POWERGEN.py` instead uses a 0.021 m^2 fixed panel, two 0.018065 m^2 wings, 0.30 cell efficiency, 0.72675 system factor, nadir-pointing attitude and a 97.4° inclined orbit. Those studies are different configurations, and their output values are not interchangeable. For an illuminated panel the basic generation calculation is

$$P_{\text{array}} = S\,\eta_{\text{cell}}\,\eta_{\text{system}} \sum_k A_k \max(0, \hat{n}_k \cdot \hat{s})\, f_{\text{deg},k}, \tag{6}$$

where S is incident solar irradiance, A_k is the panel area, $\hat{n}_k$ and $\hat{s}$ are the panel normal and Sun direction, and $f_{\text{deg},k}$ is a model degradation factor where applied. Some source studies use independently tracking wings instead of the common attitude cosine, so their specific implementation must be cited.

A.5.3 Battery Sizing and Depth of Discharge

The analytical `power/TradeCase.py` compares capacities of 10, 15, 20, 25 and 30 Wh with a fixed 97-minute period, 8 W load and a precomputed eclipse-fraction table. Its eclipse discharge estimate is

$$\text{DoD} = \frac{P_{\text{load}}\, f_{\text{eclipse}}\, T_{\text{orbit}}}{E_{\text{battery}}}. \tag{7}$$

The companion state-of-charge illustration uses 10-second steps and an assumed sunlight recharge margin of 1.3 W above the load. The study's hardcoded cycle-life figures are assumptions, not a demonstrated battery qualification. The integrated DITL uses a 25 Wh planning capacity, while the MCR also names an ICEPS2 Type A product with a 22.5 Wh specification and another table refers to 28.8 Wh. Final energy margin must be recalculated using the selected battery's usable beginning- and end-of-life capacity.

A.5.4 Mode-Resolved Electrical Loads

The main MCR gives preliminary mode-power ranges, while the integrated Science code defines its own component-specific loads, including OBC 0.8 W, EPS 1.2 W, payload 10 W and radio TX 12 W. These code inputs should not be mistaken for measured flight hardware loads. For each mode, the detailed budget should sum simultaneous loads and distinguish peak power from time-averaged energy:

$$\Delta E = \int_{t_0}^{t_1} \left(P_{\text{charge}}(t) - P_{\text{loads}}(t) \right) \, \mathrm{d}t. \tag{8}$$

The expected mode timeline, attitude-related power, battery and storage constraints must be evaluated using the same run configuration. The absence of a brownout in a simulation is evidence about that model only.

A.5.5 Science Data, Storage and Downlink

The approved preliminary link uses 19 200 bit/s, 2400 bytes/s and 100% science allocation during eligible Relay windows, with no arbitrary throughput derating. Science Record v3 occupies 292 bytes before radio framing. Gross capacity, modeled queued transfer and future verified application throughput remain distinct.

Table 21: Geometric-visit volume sensitivity; study horizon and science gating unknown

Readings/visit	Records	Bytes	Ideal Relay time (s)
1	214 039	62 499 388	26 041.412
2	428 078	124 998 776	52 082.823
3	642 117	187 498 164	78 124.235

The supplied 7940-cell table sums to 214 039 visits and 1 453 556 s geometric dwell. Its elapsed horizon is unverified; pooled dwell is not mission duration. No bytes/day normalization or annual scaling is applied. Aggregated fine-cell pass counts cannot reconstruct coarse continuous visits. This table assumes every geometric visit yields records and is not verified science production.

Table 22: Corrected annual sampled-gate workload, preferred three readings per visit

Tier	Gated visits	Records	Bytes	Peak queue
MMSC	13,291	39,873	11,642,916	20,148
MSC	39,473	118,419	34,578,348	20,148
EMSC	53,239	159,717	46,637,364	75,336

These are hypothetical record slots on sample-owned geographic cells at 5, 1 and 0.5 degrees. Exact crossing times and actual readout boundaries are unavailable. Every gated interval is retained, including zero-count exposure; partitions are nonoverlapping. Interrupted acquisition imposes a fragmentation floor: more than three segments in a visit cannot be represented as three contiguous intervals without losing exposure or bridging shutdown. This interface conflict remains open; no duration/count threshold is invented. Multiple readings are not independent orbital passages.

Table 23: Corrected annual preferred EMSC queue outage sensitivity

Removed service (days unless noted)	Peak bytes
1	211,992
3	486,180
7	1,010,028
one complete contact	146,292

Nominal EMSC peak is 75,336 bytes, compared with 262144 bytes dedicated science RAM and 134217728 bytes candidate flash. These allocations must include queue metadata and other explicitly budgeted reserves; they are not total OBC memory. The 292-byte packed assembly plus illustrative 512-byte double histogram bank uses 804 bytes before stack, DMA, operating system and other subsystems. The corrected nominal queue fits within the candidate dedicated RAM before metadata. A 16 GB archive remains an optional retention trade, not a demonstrated minimum.

The total-volume/eligible-time lower bound is 2088.061 bit/s for EMSC; it is not a guaranteed queue-closing bitrate because release timing and finite storage matter. Full-rate queue simulation transfers 46,637,364 bytes and leaves 0 bytes. Detailed ledgers retain each release, contact capacity, queue before/after, unused capacity and conserved byte balance. No physical link efficiency is inferred.

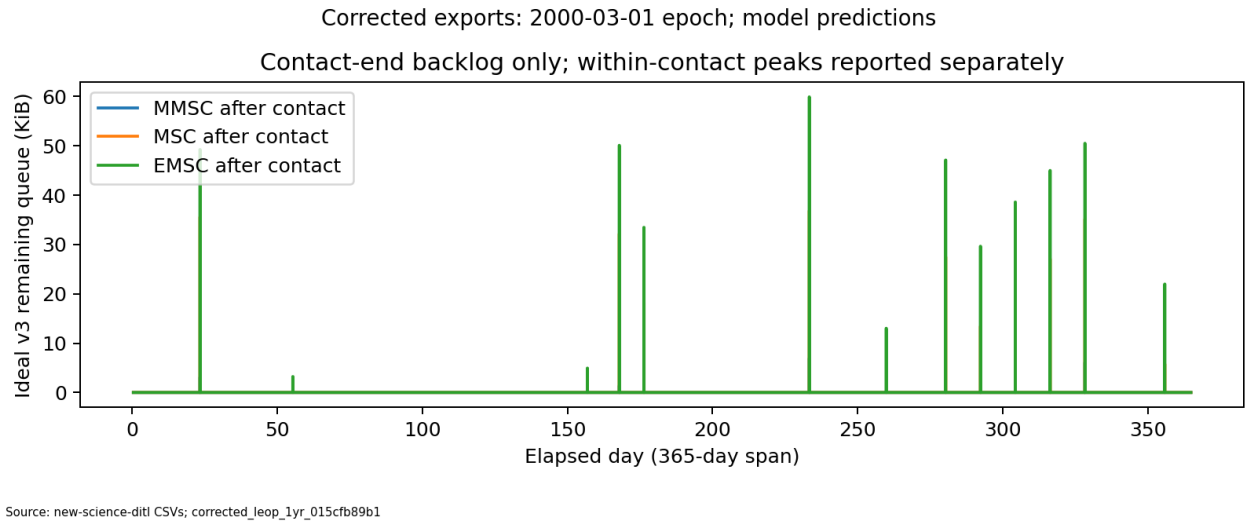


Figure 11: Corrected annual v3 sensitivity, contact-end backlog. Within-contact and pre-contact peaks are given in the tables; this curve is not emitted telemetry.

For this release/service trace, binary search gives approximately 2644.86 bit/s to clear the final backlog (unbounded storage), versus the aggregate lower bound above. The 19200 bit/s reference is 7.26 times this trace-specific value, not an RF margin. The longest no-service interval contains 64824 generated bytes.

The initial MCR records a preliminary component cost table with most entries marked TBD. No consolidated priced procurement package is established in the supplied materials. The final bill of materials will require vendor quotations, quantities, testing and integration costs, contingency, delivery lead times and ownership of custom payload manufacturing. Until these inputs are supplied the cost budget remains preliminary.

A.5.6 Recorded Energy and Budget Boundary

The corrected annual telemetry integrates 50288.809 Wh solar and 30866.103 Wh load over 365 days under the left-held convention. Recorded minimum/final battery is 15.097556/24.019245 Wh. Runtime battery capacity, charging/curtailment and logger semantics remain unverified; these integrals are not an EOL reserve margin. The independent thermal heat schedule is not this electrical history.

A.6 Orbit, Coverage and Communications Supporting Analysis

A.6.1 Orbit Propagation and Orbital Lifetime

The mission baseline is a near-circular 600 km Sun-synchronous orbit at 97.6° inclination. The supporting simulation source contains several separate orbit models rather than a single uniform propagator. The annual `power/ANTARA_MCR.py` campaign runs 365-day numerical propagation across RAAN $0^\circ, 30^\circ, \ldots, 180^\circ$ with two Monte Carlo realizations each, 300-second reporting steps, 4.0 kg spacecraft mass and 97.4° inclination. Its force model includes gravity, DTM2000 atmospheric drag, solar radiation pressure and third-body perturbations. Its SAA and ground-station encounter classifiers are simplified geographic boxes. The dedicated `orbit/ANTARA_drag_finalconfig.py` assumes a 4.0 kg spacecraft, $C_D = 2.2$, projected area 0.076799 m^2 multiplied by a 1.5 tumbling factor, 98° inclination, and the January 2000 space-weather reference. It uses DTM2000, an RK4 step of 60 seconds and samples altitude daily. It computes duration from 500–600 km initial altitudes until crossing a *450 km study cutoff*; this is not an end-to-end calculation of atmospheric reentry or post-mission disposal compliance. The newer two-run-per-RAAN drag data to a nominal 300 km cutoff are separately

Table 24: Principal standalone orbital-study differences.

Study	Key configuration	Principal limitation
Annual RAAN campaign	365 days, 7 RAAN cases, 2 runs each; 4 kg, 97.4°	Simplified SAA/GS regions; run assumptions differ from DITL
Final-configuration decay	500–600 km, DTM2000, 4 kg, 450 km cutoff	No propagation from cutoff to physical reentry
Analytical CONOPS	30 days, Eckstein–Hechler, 60 s step	Rectangular station region, not actual elevation visibility
Pointing budget	One-day Keplerian orbit, 97.4°, RAAN 0°	+Z antenna body axis, unlike nominal +Y relay

reported in Appendix H. They do not retroactively change the definitions or parameters of the earlier 450 km study. A full disposal assessment must use the applicable mission and regulatory requirements, a defined physical end criterion, atmosphere/solar-activity uncertainty, and the final deployed spacecraft configuration. One-year altitude remaining above 550 km does not by itself demonstrate compliance.

A.6.2 SAA Boundaries and Science Coverage

The fast CONOPS and annual campaign use the simplified SAA geographic region $-50^\circ \leq$ latitude $\leq 0^\circ$ and $-90^\circ \leq$ longitude $\leq 40^\circ$. This differs from the science mapping region stated elsewhere in the report. The current science region is the supplied count-rate polygon, not this historical rectangular proxy. The 30-day `conops/ANTARA_conops.py` tool uses an Eckstein–Hechler J2-perturbed orbit, 60-second samples and a rectangular Goa region defined by latitude 0–20° and longitude 60–80°. It logs eclipse and orbit classifications but does not compute true topocentric elevation. A SAA-box event, an SAA crossing, a unique orbit and an operational detector pass are different quantities and must not be described by the same count. The legacy total above 8,000 “SAA-only orbits” pools two annual runs. The recovered raw records reproduce that sum; it is not a single-spacecraft one-year revolution count. Classifier geometry remains approximate. The integrated Science simulation instead maintains contact/pass histories and counts accepted detector events and science flushes. Its

present proton-flux baseline is not a live AP9 solution; any predicted science precision must retain that limitation.

A.6.3 Ground Station Visibility and Pointing

The primary Goa station is approximately at 15°N, 74°E. The integrated Science default uses 70 m station altitude and a 15° elevation mask. The separate `pointing/ANTARA_pointing_budget.py` uses 30 m altitude and a 10° elevation mask for a one-day Keplerian case. Its antenna boresight is +Z in a nadir-pointing attitude; Science defines the relay antenna on +Y and slews to a moving ground-station line of sight. Results from the former must not be used to claim validated +Y relay acquisition in the latter. The Science code supports a second station when configured. Among simultaneously visible stations it selects the greatest elevation above each station's own mask, with primary-site tie-breaking. This is a modeled selection policy, not evidence that a second station has been selected or constructed.

A.6.4 Communications Link and Verification

The preliminary communications assessment assumes an ideal gross downlink rate of 19.2 kbps during eligible ground-station relay windows, without an arbitrary throughput derating. This establishes an upper-bound modeled data-transfer opportunity rather than verified application-layer throughput. Detailed RF link-budget closure, protocol overhead, retransmission effects, and achievable net science-data throughput are deferred to PDR.

The MCR candidate is the AX100 concept, not a verified device-rate selection. PDR must close supported bitrate/configuration, modulation/coding, EIRP/gains, pointing/polarization, path/atmospheric/implementation losses, sensitivity and E_b/N_0, link margin, framing, FEC, loss/retransmission, modem compatibility and housekeeping/command allocations. The existing elevation-dependent simulator transfer proxy is historical; it is not applied to the new full-rate capacity calculation.

Table 25: Corrected source contacts and telemetry-Relay overlap at full ideal 19.2 kbps

Scenario	Contacts	Visible s	Relay s	Ideal Relay bytes
leop_24h	2	766	646	1,550,400
no_leop_24h	2	766	646	1,550,400
leop_1yr	820	244,016	178,682	428,836,800

The source station label is GS1_India; BITS Goa is the planning station. Runtime station coordinates and elevation mask remain UNKNOWN. The producing mask is not applied again and no 10/15-degree conversion is attempted. All event timestamps identify 2000-03-01, not 2029. Native credit, ideal visibility capacity, ideal Relay capacity and future verified net throughput are distinct. Corrected per-contact and daily ledgers are in `data/derived/ditl_correction_2026-11-09/`.

$$P_{\mathrm{rx,dBm}} = P_{\mathrm{tx,dBm}} + G_{\mathrm{tx,dBi}} + G_{\mathrm{rx,dBi}} - L_{\mathrm{path,dB}} - L_{\mathrm{other,dB}}. \tag{9}$$

The final system verification should include antenna impedance matching, S-parameter characterization, antenna gain/pattern or accepted equivalent evidence, RF-chain gain and noise figure, end-to-end receive testing at the intended frequency, and downlink throughput under realistic contact conditions. No numerical link margin is claimed without the corresponding hardware and test inputs.

Historical material notice. The following appendices include transcribed earlier evidence and updated baseline/closure registers. Historical design descriptions are explicitly marked; where they conflict with the revised main sections and Appendix A, the current baseline controls. Historical figures are not current flight selections.

B Mission Baseline Register and Review Gates

This register replaces unfilled draft tables. It records approved inputs and evidence boundaries without assigning unverified margins or project dates. Historical study configurations remain individually identified in Appendices A.3 and H.

B.1 Mission Design Baseline and Assumption Register

Parameter	Current input	Qualification
Spacecraft	3U; maximum 3.0 kg	Mass budget not closed by component subtotal
Reference orbit	2029-01-01 UTC; 600 km; 97.6°; $e \sim 0.001$	Runtime/environment identity required for each result
RAAN	Launch-dependent	Representative cases are sensitivity studies, not controllable insertion choices
Battery	25 Wh planning capacity	Distinct from manufacturer pack rating
ADCS	Reaction wheels and magnetorquers	Supplier/interface selection remains open
Science	Proton differential flux, 5–100 MeV	MMSC/MSC/EMSC thresholds retained; detector and coverage validation required
Operational objective	Approximately five years	Separate from modeled 300 km cutoff and reentry

B.2 Mission-Level Design Trades

The current payload configuration is controlled by Appendix A.2. Radiation input dates and RAAN evidence identities are controlled by Appendix A.3.8. EPS, antenna and station studies retain their own configuration assumptions; a common spacecraft feasibility claim requires consistent loads, geometry, exposure and data products. No component selection or physics model is changed merely to reconcile a table.

B.3 Development Gates

PDR requires controlled payload CAD/optics, acquisition and response-validation plans, science eligibility/independence rules and closed interface ownership. CDR requires implemented and verified hardware/software against those baselines. Environmental qualification and flight readiness require documented component/assembly tests and launch interfaces. Dates and named owners require team assignment; this report does not invent schedule commitments.

C Source Subsystem Detail and Tables

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

The ANTARA spacecraft architecture has been developed as a modular, scalable, and mission-oriented CubeSat platform optimised for scientific radiation measurements withi n the South Atlantic Anomaly. The spacecraft follows a centralised avionics architecture in which the On-Board Computer (OBC) functions as the primary command. The communications subsystem, Payload Subsystem and ADCS subsystem work on the basis of the instantaneous mode. Following is the spacecraft system architecture.

C.1 System Architecture Overview

The ANTARA spacecraft is a **3U CubeSat**, conforming to the standard 3U form factor of 100 × 100 × 340.5 mm as defined by the CubeSat Design Specification (CDS) within the standard 3U mass allowance. Compliance with the applicable CDS revision, including deployer-imposed mass, center-of-gravity, and rail/standoff constraints, will be formally verified once a launch integrator and deployer are selected.

The five primary spacecraft subsystems integrated around a centralised avionics architecture are: On-Board Computer (OBC), Electrical Power System (EPS), Attitude Determination and Control System (ADCS), Communications System (COMMS), and Scientific Payload.

C.1.1 Electrical Power System (EPS)

The Electrical Power Subsystem is responsible for generating, storing, and distributing electrical power to all spacecraft loads. The COTS model we have taken for the simulation is the ISISPACE ICEPS2 Type A compact EPS, which includes an integrated 2-cell battery pack, autonomous battery heating, I2C communication, up to 6 output voltage domains, and protection features including overcurrent and reverse-current protection.

Table 26: ICEPS2 Type A parameters

Parameter	Preliminary Requirement
Mass	184 ± 5 g
Volume	96 × 92 × 26.45 mm
Energy storage	22.5 Wh (Type A specification)
Operating temperature	−20°C to +70°C
Interfaces	I2C housekeeping/control
Voltage domains	Regulated 3.3 V, 5 V, unregulated bus; up to 6 output domains

Note: The mission simulation uses a 25 Wh battery model as a conservative planning estimate. The final power budget must be reconciled against the ICEPS2 Type A 22.5 Wh specification at PDR.

Mode Power Estimates:

Table 27: Spacecraft operating-mode power estimates

Mode	Active Components	Estimated Power (W)
Cold & Dark Boot	EPS + OBC	0.8
Detumbling	EPS + OBC + ADCS MTQs	3–6
Safe Mode	EPS + OBC + Radio RX	1–2
Sun-Pointing	EPS + OBC + ADCS	2–3
Detection Mode	EPS + OBC + Payload + ADCS	6–9
Relay Mode	EPS + OBC + Radio TX + ADCS	4–7

Table 28: Preliminary subsystem electrical power budget

Subsystem	Idle / Standby (W)	Nominal (W)	Peak (W)
EPS	-0.093	0	0
Battery	-	charging capacity(28.8Wh)	-
OBC	-	-0.17	-0.9
Radio	-0.15 (rx)	-2.64 (tx)	-3 (TX max)
Unidirectional antenna	-0.04	-0.06	-2 (Deployment)
ADCS	-0.8	-2.6	-4.5
Payload (Assumed)	-	-3.0	-5.0
Solar Panels	-	11.5W	22W
GNSS Navigation	-0.21W	-0.255W	-0.39W

*Payload data is yet to be obtained, Worse case scenarios for each state is taken

C.1.2 Communications System (COMMS)

The Communications System provides the spacecraft's interface with the ground segment and supports telemetry transmission etc. The COTS model we have taken for the simulation is the GomSpace NanoCom AX100 radio system working on two antenna systems: GomSpace ANT430 omnidirectional antenna and ISIS UHF dipole antenna.

The primary ground station is planned for **Goa, India**.

Table 29: Goa ground station assumptions

Parameter	Value
Latitude	15°N
Longitude	74°E
Altitude	70 m
Elevation mask	15°

Simulation predicts an average of **two contacts per day**, with durations of approximately **368 s** and **414 s**. The maximum elevations are approximately **38.5°** and **77.5°**, providing roughly **780 s of total daily contact time**.

Although our simulations currently have one ground station, we aim to have more ground stations for receiving purposes.

Communications Concept

ANTARA uses a **Relay Mode** architecture, with communications restricted to ground-station

contact periods. During Relay Mode, the spacecraft performs:

- Telemetry transmission
- Science-data downlink
- Command reception
- Operational synchronisation

Downlink Architecture

The baseline downlink rate is **15–18 kbps (historical, superseded by the ideal 19.2 kbps MCR assumption)**.

The scientific payload does not transmit individual proton events. Instead, the onboard FPGA accumulates events into **64-bin spatial-energy histograms**. Each histogram contains:

64×4=256 bytes

At one update per minute, the science data rate is approximately:

(256×8)/60≈34 bps

This corresponds to an estimated **300–400 KB/day (historical estimate, superseded by the v3 sensitivity analysis)** of science data.

With two daily contacts totalling approximately **780 s**, a 15 kbps link provides:

15,000×780≈11.7 Mbit

Thus, the available capacity provides approximately a 4× margin over the expected science-data volume, allowing for housekeeping, operational logs, attitude data, protocol overhead, and retransmissions.

The **9.6 kbps** value used in earlier analysis is superseded by the architecture-driven **15–18 kbps (historical, superseded by the ideal 19.2 kbps MCR assumption) baseline**.

C.1.3 Historical Payload Subsystem

Superseded architecture. The silicon entrance, five-layer stopping telescope, rear veto, EJ-200 selection and calibrated-bin language below are historical only. Current construction and response limitations are in Appendix A.2.

Overview

The payload is a **Flux Particle Detector Unit (FPDU)**: a compact proton spectrometer using a silicon ΔE front detector followed by a 5-layer EJ-200 plastic scintillator stack, read out by SiPMs, with a rear veto layer. It measures trapped protons from **5–100 MeV** in **64 log-spaced energy bins** at Sun-synchronous orbit (~600 km)

Physical Stack — Dimensions Along Boresight

Table 30: FPDU telescope physical-stack dimensions

Component	Thickness	Notes
Aperture tube + structure	30 mm	Sets θ_max = 18.4°
Si wafer	1 mm	Active detector
Si mount + PCB	3 mm	Carrier board
L1 + SiPM + coupling	4 mm	Scintillator** + SiPM package
L2 + SiPM + coupling	6 mm	
L3 + SiPM + coupling	14 mm	
L4 + SiPM + coupling	24 mm	
L5 + SiPM + coupling	32 mm	
Rear veto + SiPM	4 mm	
Optical isolation gaps	~3 mm	Black wrap between layers
Readout electronics board	15 mm	FPGA, ADCs, shaping, bias
Structural margin	5 mm	
Total	**~141 mm**	Fits in 2U payload volume

****All these calculations are done assuming EJ200 plastic scintillator and is subject to change.**

Dead Time and Livetime

Table 31: FPDU dead-time and livetime assumptions

Parameter	Value
Dead time per event	**~250 ns (shaping + ADC readout)**
Peak total event rate (SAA core)	**~100 Hz**
Dead time fraction at peak flux	**12.5%**
Livetime fraction at peak flux	**87.5%**
Dead time at SAA periphery	**< 0.1% (negligible)**

C.1.4 Historical Attitude Determination and Control System (ADCS)

Superseded selection. The CMG description below is not the current reaction-wheel and magnetorquer concept.

Attitude Determination and Control System(ADCS) continuously processes sensor data to calculate the spacecraft's exact position relative to external reference points. The COTS model we have taken for the simulation is TensorTech ADCS - 10m - D. The model uses a comprehensive suite of components: one ADCS-Main Control Board (ADCS-MCB), four TensorFSS-15M Fine Sun Sensors, and one integrated Control Moment Gyroscope (CMG) model.

Table 32: TensorTech ADCS-10m-D preliminary characteristics

Characteristic	Value
Pointing Knowledge	0.3 degree (general) 3-100 arcseconds (with startracker)
Pointing Accuracy	0.4 degree
Power consumption	0.8W (idle) 2.6W (nominal) 4.5W (Max)
Mass	450g

Control Modes and Algorithms

Detumbling. Following deployment, the ADCS executes a two-stage control law: an initial B-dot magnetic detumble to reduce high angular rates acquired during deployment, followed by reaction-wheel damping to bring the spacecraft to a stable, low-rate condition suitable for Sun Pointing Mode entry.

Simulation results show detumble completing in approximately 2,460 seconds from initial body rates of [6.0, −5.5, 4.5] °/s. No hard detumble timeout is currently defined.

Sun Pointing. The ADCS orients the spacecraft +Z face toward the Sun to maximize solar-array power generation, using reaction wheels for fine pointing and magnetorquers for momentum dumping.

Detection Mode Pointing. During SAA traversal, the ADCS aligns the +X axis (payload boresight) with the spacecraft velocity vector to support directional proton flux measurement and East-West asymmetry analysis. Simulation results show a mean attitude error of approximately 2°, against a threshold of $\leq$5° and a goal of $\leq$2°.

Relay Mode Pointing. The ADCS points the +Y face (directional relay antenna) toward the ground station to optimize downlink signal strength during scheduled contacts.

Slewing. Mode transitions between Sun Pointing, Detection, and Relay require dedicated slew maneuvers. The 24-hour DITL simulation recorded 18 slew events with slew angles ranging from 5.4° to 175.8°; large-angle slews were identified as consuming non-negligible time and should be factored into mission scheduling and duty-cycle analysis.

Saturation Handling. Reaction wheel momentum saturation is managed through magnetorquer-based momentum unloading. If saturation persists beyond a defined hold timeout and pointing diverges, the FSM commands a transition to Safe Mode as a fault-protection response, this is the second-highest-priority transition in the mode state machine after critical battery conditions

Table 33: ADCS simulation performance summary

Metric	Requirement / Result
Detumble completion	~2,460 s from [6.0, −5.5, 4.5] °/s
Detection Mode pointing error	<2°
Slew events (24-hr sim)	18
Slew angle range	5.4°–175.8°
Momentum management	Magnetorquer unloading + Safe Mode fallback

Pointing Budget summary:

Ground station contact pointing budget analysis is being mentioned. The result is for a sim-

ulation taken to Propagate 1-day Keplerian orbit (analytical, no perturbations) with nadir-pointing attitude, computes off-boresight angle between +Z body axis (antenna boresight) and ground station LOS vector during passes above 10° elevation mask. Outputs pass statistics and off-boresight angle plots.

C.1.5 On-Board Computer (OBC)

The On-Board Computer is the central computational and operational-control unit of the ANTARA spacecraft. The COTS model we have taken for the simulation is GomSpace Nanomind A3200.

Purpose

The OBC serves as the spacecraft's primary operational-management system and continuously monitors spacecraft status while coordinating all major mission activities.

At 3.3V bus channel, The OBC can run on various clock speeds from 8 - 32MHz depending on the state of the CubeSat with an average power consumption of 0.17W

Functional Responsibilities

The ANTARA OBC performs mission sequencing, state monitoring, command execution, health monitoring, payload control, data-storage management, and communications scheduling. Payload control is closely integrated with orbit propagation, ADCS orientation, SAA traversal prediction, and spacecraft power availability, enabling synchronised science operations during South Atlantic Anomaly crossings.

Flight Software Philosophy

ANTARA uses a state-machine-based flight software architecture designed to maximise reliability, deterministic behaviour, operational predictability, fault containment, and verification simplicity. Each operational mode corresponds to a defined spacecraft state. State transitions occur only when predefined operational conditions are satisfied, such as battery thresholds, spacecraft attitude stability, payload readiness, and subsystem health. The state-machine architecture simplifies mission simulation, operational verification, fault diagnosis, and long-duration autonomous operations.

C.2 Spacecraft Coordinate System

A spacecraft-fixed body-frame coordinate system is used throughout ANTARA mission operations. The body-frame definition is designed around payload operational geometry, solar-panel placement, communications requirements, and CubeSat deployment constraints.

Table 34: Spacecraft body-frame configuration

Axis	Function
+X	RAM face / payload boresight — primary payload pointing direction during Detection Mode
+Y	Directional relay antenna face (nominally nadir-pointing in default hold; pointed to ground station during Relay Mode)
+Z	Fixed solar panel face (pointed toward Sun during Sun Pointing Mode)
-Y	Deployable solar panel packed face
-Z	Omnidirectional communications antenna face (Safe Mode / beacon transmission)
-X	PPOD deployment face

C.3 Ground Segment & Mission Operations

Ground Station Hardware

Primary ground station: Goa, India (15°N, 74°E, 70 m altitude, 15° elevation mask).

Hardware baseline:

- **Antenna:** UHF Yagi or crossed-Yagi on an az-el tracking mount, matched to the NanoCom AX100 downlink band
- **RF front-end:** transceiver/SDR compatible with AX100 modulation, LNA at the antenna feed to minimize cable-loss impact on receive sensitivity
- **Backup/contingency station:** none currently defined, remains an open item.

Secondary Ground Station: We are yet to determine secondary ground stations. An ideal ground station would be at a polar location to accommodate for the SSO orbit. A temporary simulation was taken for Svalbard (78°N, 15°E, 50 m altitude, 5° elevation mask).

Ground Station Software

- **Orbit propagation & tracking:** TLE-driven antenna pointing; TLEs refreshed at minimum daily from CelesTrak/Space-Track, more frequently in the days immediately following deployment when orbit uncertainty is highest
- **Pass prediction:** AOS/LOS scheduling consistent with the two-contacts-per-day, ~780 s total daily contact time
- **Command uplink interface:** command construction, queuing, and uplink receipt confirmation
- Documented shift handover: spacecraft state, open anomalies, and next-pass command queue passed between operators

D Superseded Source Integrated Simulation Detail

SUPERSEDED EVIDENCE: every numerical result and figure in this appendix is retained solely as an original-draft archive. None supports current requirements, budgets or feasibility. Corrected findings are in Section 4.1 and Appendix F.

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

The Day-in-the-Life (DITL) Simulation represents one of the most important mission-analysis and systems-engineering activities within the ANTARA mission framework. The DITL simulation acts as an integrated operational verification environment used to evaluate whether the spacecraft can successfully achieve its scientific objectives while operating within the physical, operational, and resource constraints of a 3U CubeSat platform.

Unlike isolated subsystem-level analysis, the DITL framework evaluates the spacecraft as a complete operational system in which orbit dynamics, spacecraft attitude, power generation, battery charging and discharge, payload operations, ground-station communications, and mission-mode transitions all interact simultaneously.

The baseline DITL simulation duration for Mission Concept Review activities is approximately 12 hours. The team has additionally completed a year-long simulation using Orekit and Python-based subsystem models.

The DITL simulation tracks the following operational metrics:

- spacecraft position, velocity, latitude, longitude, altitude, beta angle, eclipse status
- Battery energy and SOC, solar array power generation, spacecraft electrical load consumption, net power balance
- Attitude error, pointing accuracy, reaction wheel momentum, slew events
- SAA detection opportunities, science data volume generated
- Ground-station visibility, contact timing and duration, and cumulative data-return volume

.

Executive takeaways

- Operationally stable. The spacecraft completes startup detumble, repeated science and relay activities, and returns to SUN_POINTING after each activity. SUN_POINTING occupies 87.5% of the run.
- Attitude control is accurate outside slews. Steady-state 95th-percentile pointing and attitude error are both 0.4 deg. The larger error peaks are associated with commanded reorientation events.
- Slew execution is well matched to the model. There are 18 slews totaling 18.8 minutes. Actual duration differs from the estimate by only 0.02 s on average, with a maximum difference of 0.31 s.

Radiation operations reduce science availability

The run includes six SAA passes totaling approximately 44.9 minutes.

Across the five proton passes

- Historical detection-conditional sample fraction is approximately 62.7%; not SAA duration-weighted duty
- The lowest science-active fraction is approximately 37.9%
- Other passes generally remain between approximately 66% and 71%

This suggests that the radiation protection logic is active but does not completely disable science collection during all proton environment.

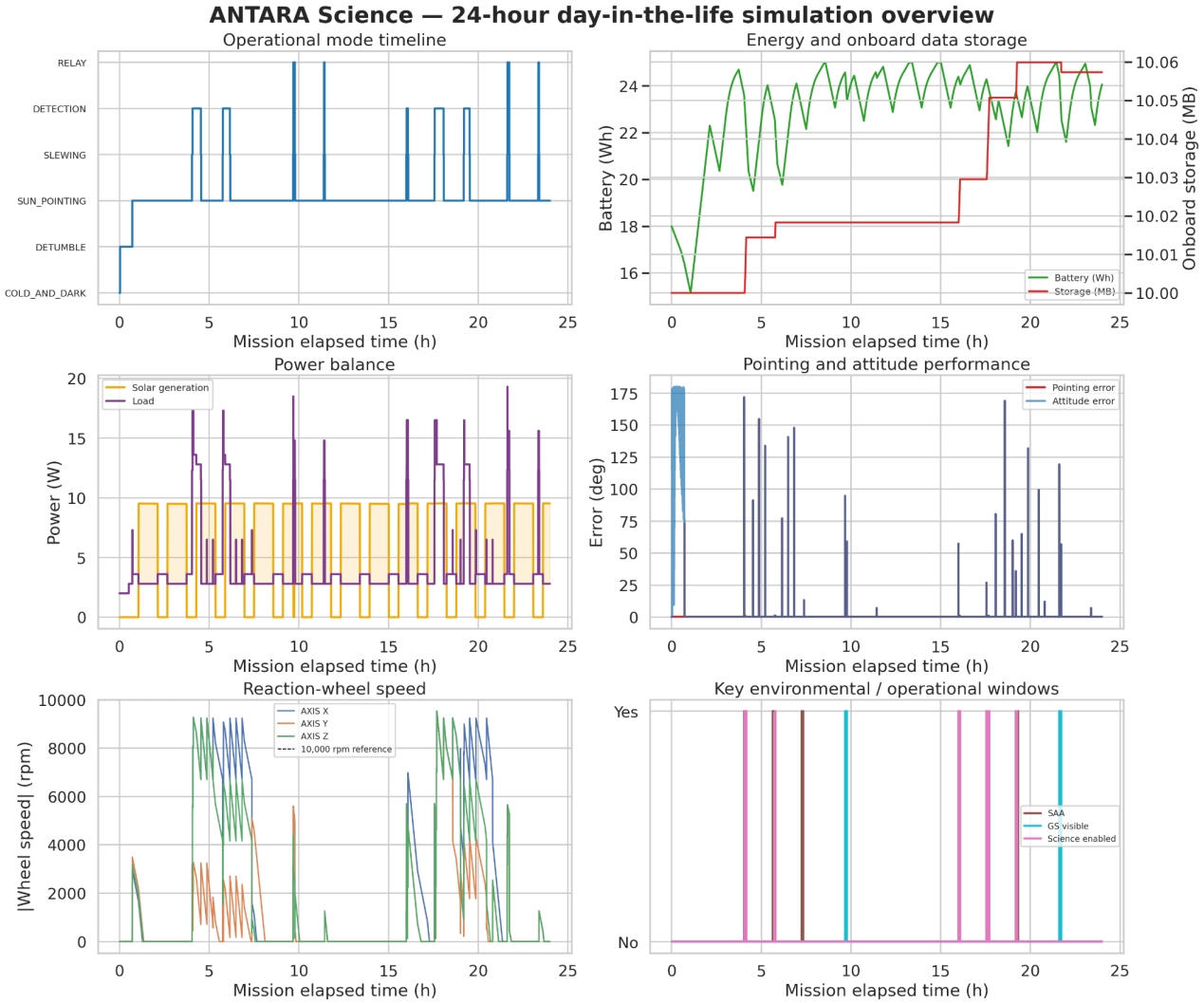


Figure 12: Superseded historical source: 24-hour integrated DITL overview: mode timeline, energy and storage, power, pointing, reaction-wheel speed and operational windows.

The battery never approaches depletion during this run.

- Initial battery energy is 18.00 Wh
- Minimum battery energy is 15.15 Wh
- Minimum level is approximately 84.2% of the initial value

- Final battery energy is 24.06 Wh
- Net battery gain over the simulation is approximately 6.06 Wh

Operational Mode Distribution (24 hours):

Table 35: Superseded historical source: 24-hour operational-mode distribution

Mode	Time Steps	Fraction of Day
Sun Pointing	37,781	87.5%
Detection	3,134	7.3%
Detumble	1,230	2.8%
Slewing	545	1.3%
Relay	450	1.0%
Cold & Dark	61	0.1%

Power and Energy:

- Average solar generation: 6.05 W
- Minimum battery energy recorded: 15.15 Wh (60.6% of 25 Wh planning capacity)
- No brownouts observed; battery remained above 30% safety threshold throughout the full 24-hour simulation.

SAA Operations:

- 6 SAA passes observed in 24 hours
- 5 of 6 passes achieved active payload science collection
- Total SAA dwell time: <2,700 seconds over 24 hours
- Science data generated: 0.060 MB (60 KB) per 24-hour period, consistent with 300–400 KB/day (historical estimate, superseded by the v3 sensitivity analysis) estimate at full nominal operations.

Ground Station Contact:

- 2 contacts with GS1_India (Goa, 15°N, 74°E) in the 24-hour window
- Contact 1: 368 seconds, maximum elevation 38.5°
- Contact 2: 414 seconds, maximum elevation 77.5°
- Daily downlink capacity at 15–18 kbps (historical, superseded by the ideal 19.2 kbps MCR assumption): ~1.5 MB/day — approximately 4× the expected daily science data volume.

ADCS Performance:

- Detumble completed in 2,460 seconds from initial body rates of [6.0, −5.5, 4.5] °/s
- 18 slew events recorded in 24 hours
- The raw 18-event table gives a minimum slew of 0.442496° (the earlier 5.4° minimum is superseded); the reported maximum is 175.8°; large slews indicate some mode transitions require significant attitude manoeuvre time and should be factored into operational scheduling.

Proton Detection (5 active science passes):

- Historical synthetic FPDU proxy: 50–70% of the configured AP9-named baseline, not measured detector performance or live AP9 across all energy channels (5–100 MeV), consistent with partial SAA coverage and pitch-angle sampling effects.
- Mean pitch angle ranged 15.6°–77.6° across passes
- Withdrawn count/precision claim: the historical 13,000–108,000 values summed rates without the required time integration. They cannot establish counts or mission-tier precision.

One-Year Extended Simulation (RAAN 180° baseline):

Table 36: Superseded historical source: Historical standalone campaign statistics; not verified annual integrated Science output

Metric	Mean	Min	Max	99% CI
Average Power (W)	11.22	11.19	11.25	11.16–11.27
Average SOC	0.987	0.986	0.987	0.986–0.987
Minimum SOC	0.834	0.834	0.834	0.834–0.834
Battery Cycles	787	785	789	783–791

E Engineering Closure and PDR Entry Conditions

The MCR establishes a conditional concept and an explicit verification path. The former placeholder-only tables in this appendix are replaced by the evidence dispositions below. No subsystem is assigned a technology-readiness level or margin without a documented basis.

E.1 Budget Evidence and Limits

Budget	Current evidence	Closure condition
Mass	3.0 kg design maximum; 1536.3 g in supplied component entries	Add payload, harness, interfaces and verified quantities; Appendix G.7
Battery	25 Wh simulation planning capacity	Common load/array configuration and accepted-energy throughput; Appendix A.5
Power	RAAN-dependent preliminary predictions	Current electronics load and EOL conditions; no closed FPDU power budget
Science / downlink	Provisional 64-bin logical record	Encoding, cadence, loss/quality metadata, storage and scheduled link throughput
Payload packaging	72 mm active length, 2.592 cm^3 scintillator	Controlled CAD including gaps, coating, sensors, shield and housing
Procurement	Candidate component evidence only	Selected part numbers, interface compatibility and quotations

E.2 Verification and Maturation

Payload qualification requires geometry-matched transport, optics and mixed-species calibration; selected electronics must demonstrate timing, throughput, buffering and measured dead time. Software tests establish accounting only. Integrated acceptance requires runtime/environment manifests, consistent exposure, complete horizons and verified data delivery. Accepted 2026 radiation and RAAN references retain their original identities.

E.3 Open Decisions and Review Gates

Payload material, optical isolation, front treatment, optional rear detector, sensor/ASIC/ADC mapping and MCU/FPGA selection remain open. Science owners must approve eligibility, independence and uncertainty definitions while retaining the committed MMSC/MSC/EMSC thresholds. Assignment of named owners and dates belongs to the project review; none is fabricated here. The repository's remaining-decisions register lists the evidence needed for each closure.

E.4 MCR Exit and PDR Entry

The MCR establishes a concept, reproducible idealized rate calculations and tested accounting, not calibrated payload performance or mission success. PDR requires controlled interfaces, acquisition timing/throughput evidence, a response campaign and science-coverage acceptance rules. Apply and retest the delivered source patch before new coupled predictions. Long campaigns require separate authorization after bounded verification.

F Integrated Mission Simulation and DITL Supporting Analysis

F.1 Purpose and Source

The integrated model in `src/simulations/ditl/ANTARA_Science_Sim.py` combines orbit propagation, spacecraft attitude, electrical power, mode transitions, proton-environment assumptions and detector accumulation, and ground-station communications. Its current implementation is a simulation model, not a hardware-in-the-loop qualification system. The shared configuration and individual presets are documented in Appendix A.3.

F.2 Defined Mission Scenarios

The source defines `leop_24h`, `no_leop_24h` and `leop_1yr`. The first includes Cold and Dark and Detumble; the second begins in Sun Pointing without LEOP startup; the annual scenario includes startup over 8,760 hours. Command-line values may override preset defaults. A comparison of the three scenarios is meaningful only when each output is accompanied by its exact version and configuration.

F.3 Mode Execution, Time and Energy

During startup, detumble and slewing, the simulator uses its fine time cadence. Other operating modes may use a coarser step. Mode occupancy, energy use, detector accumulation and mission elapsed time advance with the same actual integration interval. The corrected exports lack interval-end columns; this audit uses successive timestamps and excludes the endpoint row duration. Explicit interval fields in newer source are not retroactively attributed to these files. Reported fractions should therefore use total elapsed time, not raw row fractions. The ADCS model retains finite slew and settling requirements, momentum saturation handling and recovery logic. During saturation hold, scientific collection and downlink transmission are inhibited in the documented model. The end of a planned slew duration is not by itself proof of achieved target alignment.

F.4 Science Data, Detector Statistics and Model Limits

The detector model accumulates expected bin counts and serializes histograms into science data separately. An interval with accepted events need not produce a positive new packet volume until a flush. SAA-exit and simulation-horizon flushes belong to the corresponding pass totals. Bin precision estimates use $1/\sqrt{N}$ for accumulated expected counts, but this does not represent calibrated detector response, stochastic-count realization or a fully validated low-count confidence interval. The environment model's legacy AP9-named variable is a configured baseline and not an independent AP9 computation.

F.5 Corrected Run Identities and Accounting

The authoritative registry binds all 24 files to SHA-256 identities. The corrected daily telemetry contains 7696 and 6553 rows; annual telemetry contains 1434872 rows. All begin at elapsed zero and include the named endpoint. The annual span is exactly 365 days, from 2000-03-01 to 2001-03-01 UTC; the leap day precedes the start. This establishes timestamp coverage, not the missing runtime configuration or hardware validity. The former 50-day truncation finding applies only to superseded inputs.

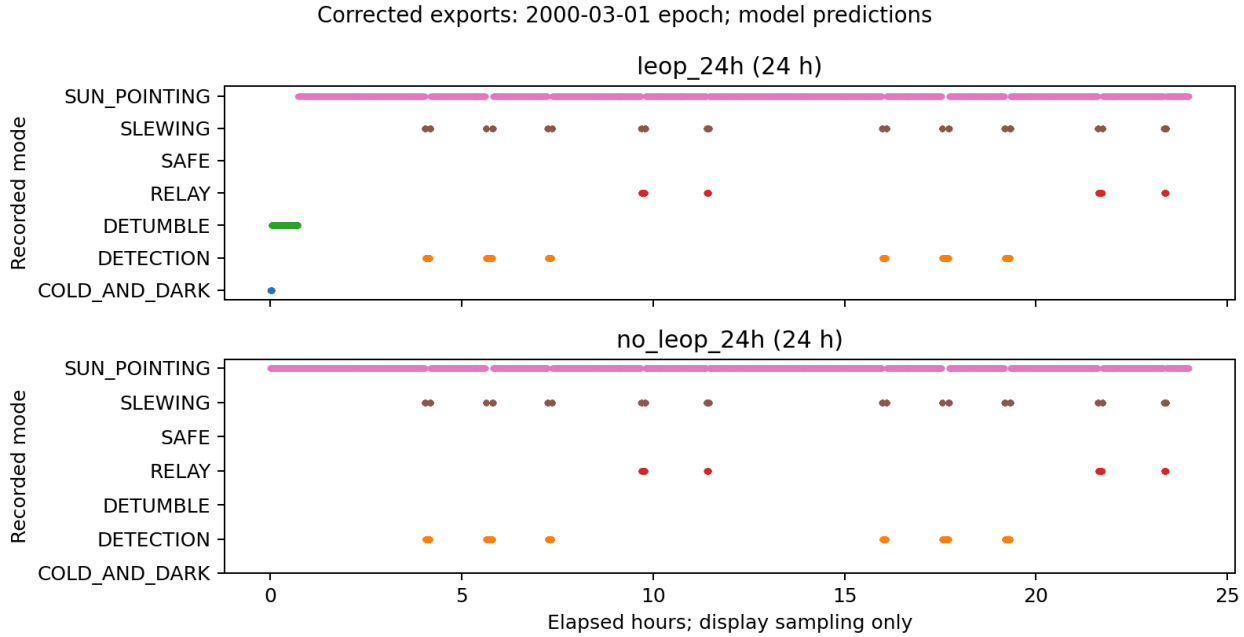
No duplicate or nonmonotonic telemetry timestamps were found. Day-run timestamp dif-

ferences are 2/20 s. Annual differences are 2/60 s with one 50 s interval, distinct from the current source default. Annual telemetry has 15161 mode changes versus 16283 transition records and 425 sampled Safe entries. Exact event checks are retained in the JSON audit. The expected adaptive schedule is unknown, so elapsed coverage alone cannot certify that every intended sample was exported. The last sample contributes no additional duration; its native increment fields remain included in cumulative data checks.

Table 37: Transition-log versus left-held telemetry mode duration (annual, seconds)

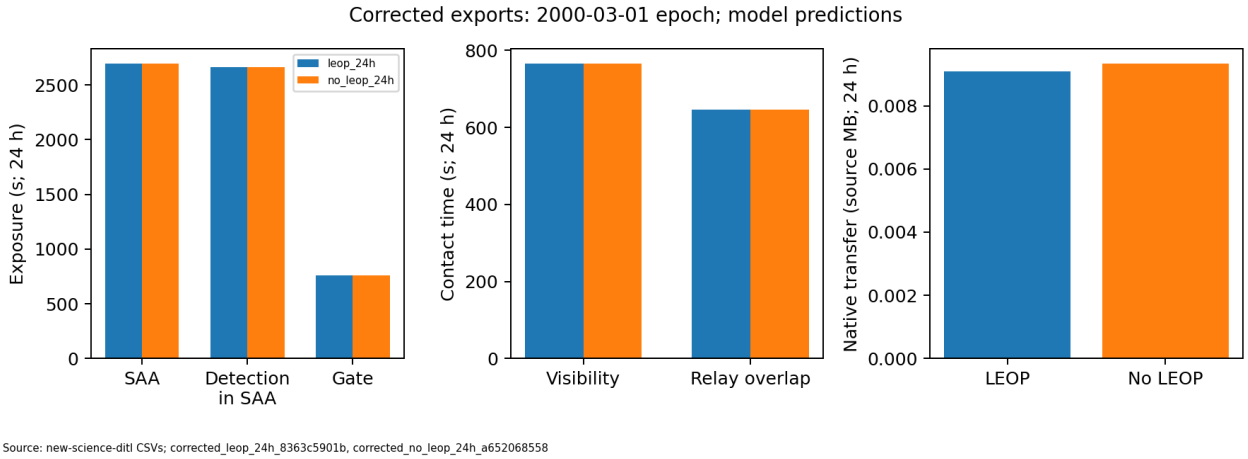
Mode	Telemetry	Transitions
COLD_AND_DARK	122	120
DETECTION	894,054	889,226
DETUMBLE	2,460	2,460
RELAY	269,876	266,776
SAFE	332,882	331,350
SLEWING	375,688	620,804
SUN_POINTING	29,660,918	29,425,264

Transition mode sequence matches: False; timestamp offset range is undefined because the sequences differ. Slew events total 512,586 s across 7928 rows, distinct from 375,688 s Slewing occupancy. These differences require the generating logger/control semantics, not fabricated interval adjustments.



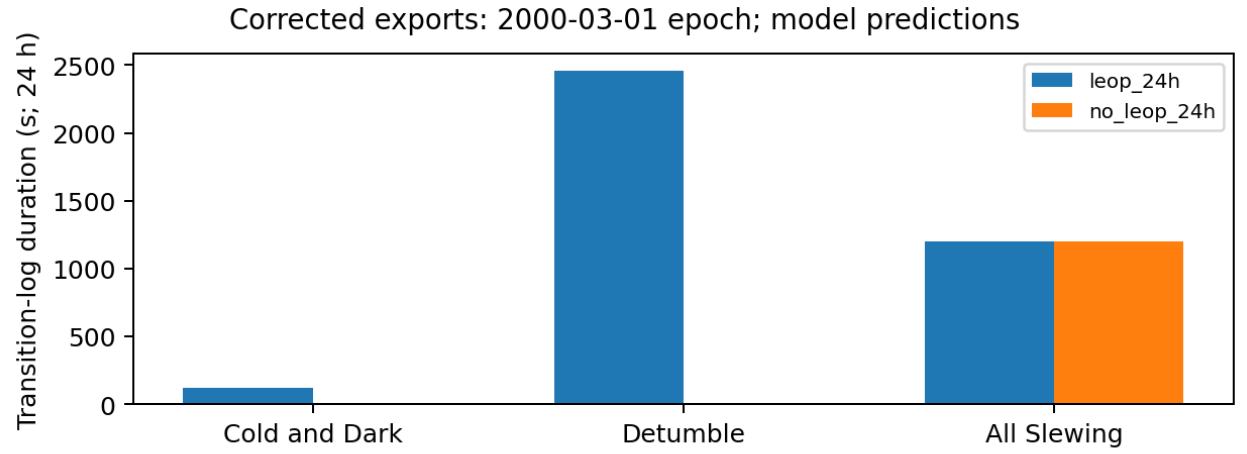
Source: new-science-ditl CSVs; corrected_leop_24h_8363c5901b, corrected_no_leop_24h_a652068558

Figure 13: Corrected 24-hour mode timelines. Display-sampled points do not define the integration weights.



Source: new-science-ditl CSVs; corrected_leop_24h_8363c5901b, corrected_no_leop_24h_a652068558

Figure 14: Exposure gates, eligible contact time and native transfer credits for the corrected day runs. Credits are not confirmed ground receipt.



Source: new-science-ditl CSVs; corrected_leop_24h_8363c5901b, corrected_no_leop_24h_a652068558

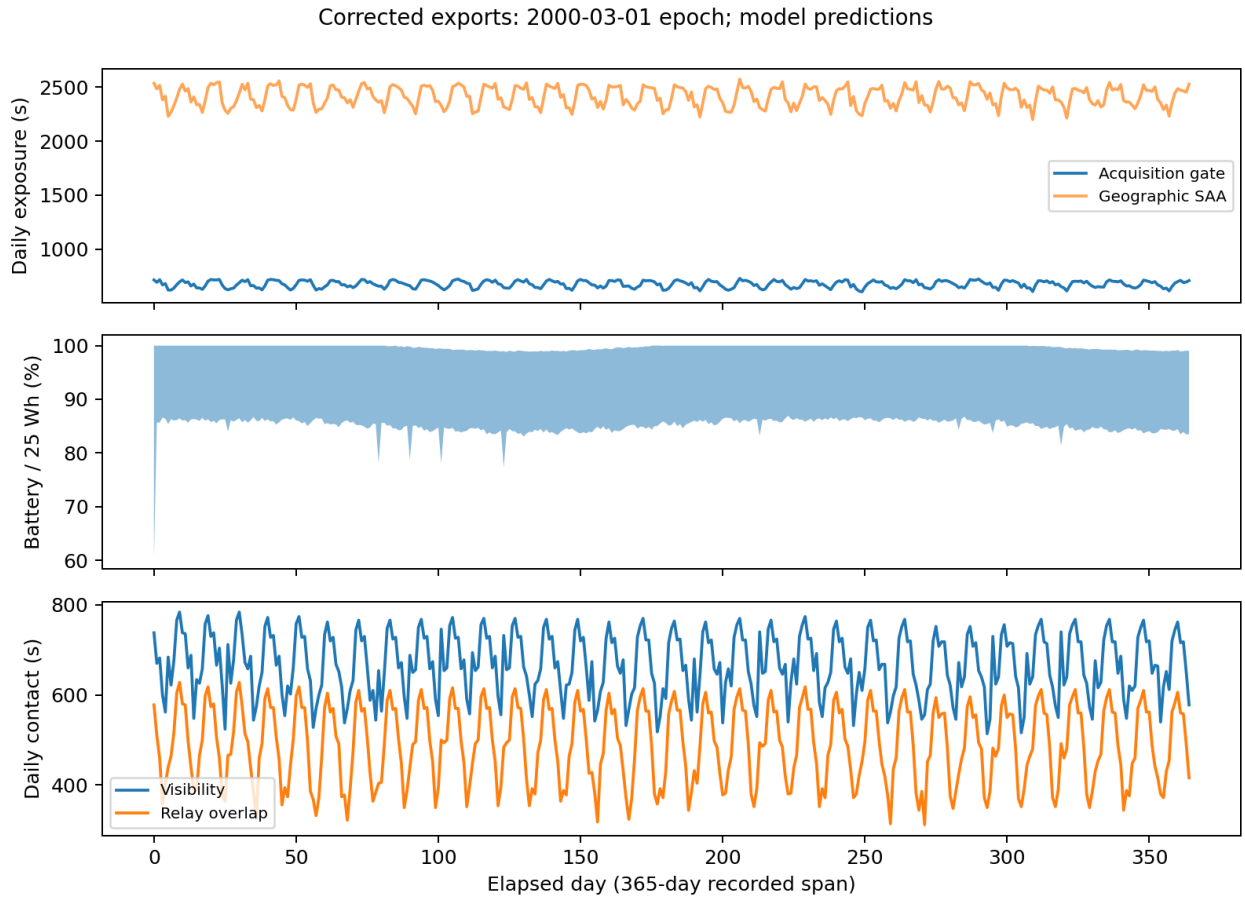
Figure 15: Startup and Slewing duration from transition logs, shown separately from telemetry occupancy.

F.6 Annual Exposure, Energy and Native Data

There are 2346 SAA event records and 2346 sampled SAA windows. Independent science passages additionally require the approved eligibility and quality contract. The acquisition gate totals 247,258 s. Per-pass array sums and telemetry gate-array sums are 3.10842e+09 and 3.10842e+09; these are legacy model expectations, not reconstructed current-instrument events. Bin-wise rate-times-duration comparisons can differ after merging; effective channel metadata and dimensional calibration remain open.

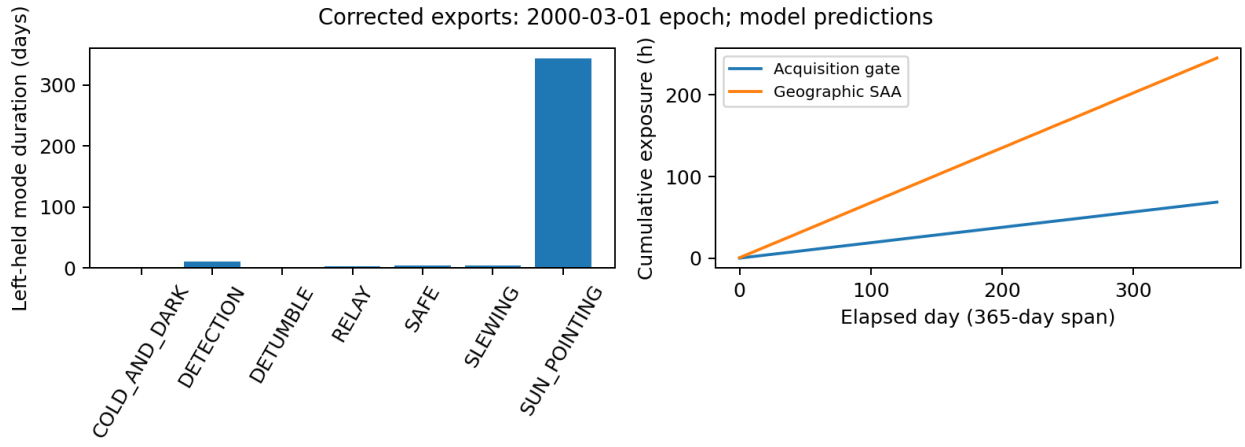
Recorded battery spans 15.097556–25.000000 Wh, ending at 24.019245 Wh. No negative battery row was found. The solar/load integrals are 50288.809/30866.103 Wh under the stated interval convention. Charge efficiency, curtailment and post-step battery semantics prevent treating their difference as stored-energy margin or EOL feasibility.

Native store begins at 10, peaks at 10.005966187, and ends at 10 source MB. Generation minus transfer closes against storage to 7.64e-14 source MB. Histogram-byte totals do not alone establish packet serialization completeness. Raw event packets, lost-event counters and ground-reception archives are absent.



Source: new-science-ditl CSVs; corrected_leop_1yr_015cfb89b1

Figure 16: Corrected 365-day exposure, battery envelope and contact availability; epoch 2000-03-01. These are recorded model predictions under unresolved runtime configuration.



Source: new-science-ditl CSVs; corrected_leop_1yr_015cfb89b1

Figure 17: Annual telemetry mode occupancy and cumulative geographic versus acquisition-gate exposure. No mission-success classification is inferred.

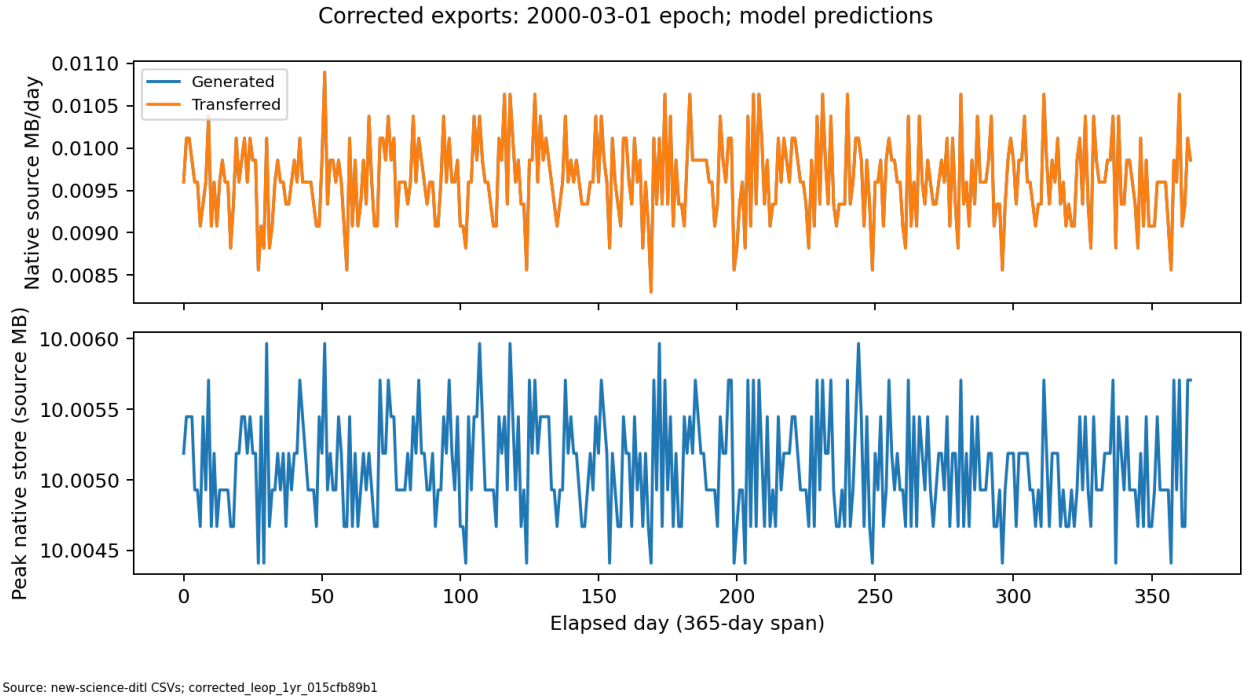


Figure 18: Daily native generation/transfer and maximum source-labelled store. The historical native packet format is not Science Record v3.

F.7 Historical Detumble Summary

The supplied 50 000-row summary reports 16 133 successful trials and 33 867 nonconverged trials; all unsuccessful completion times are infinite. Successful-only median, 95th percentile and maximum times are 951.5, 2569.0 and 3026.5 s. Whole-population convergence quantiles and flight reliability are not established. The first baseline row completes in 1009 s, distinct from the integrated 2460 s startup. Initial spin magnitudes range from approximately 0.0008 to 60 deg/s. All rows specify nonzero initial accelerations; the current source rejects that independent-acceleration model, preventing exact reproduction with current code.

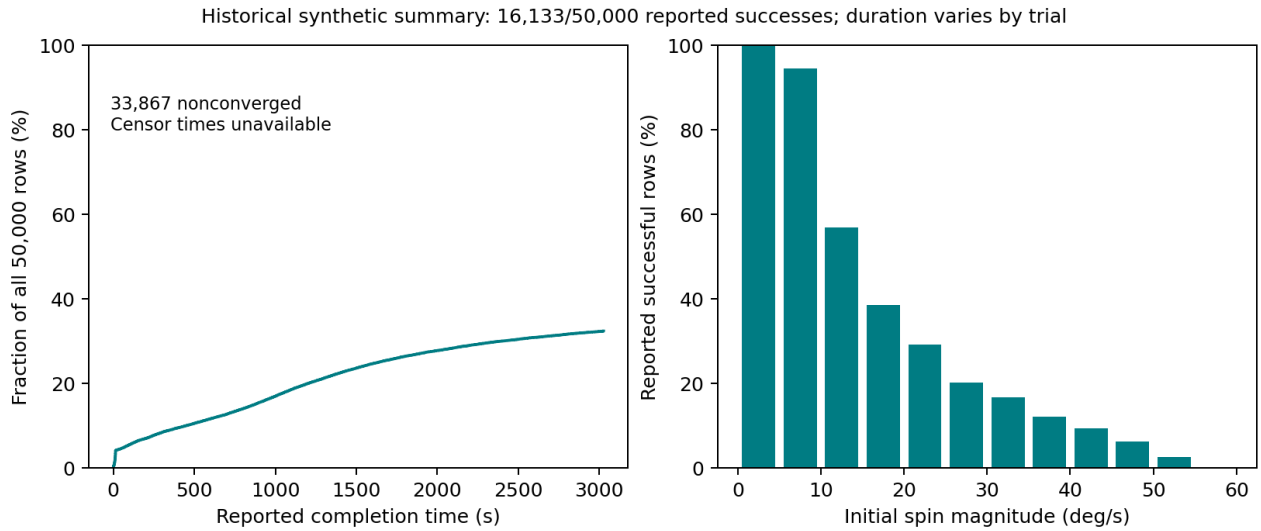


Figure 19: Historical detumble summary. The empirical completion curve retains all 50 000 rows as denominator; it is not a censor-adjusted survival estimate. Spin-bin proportions do not establish flight initial-condition probabilities.

The summary marks 25 473 rate-cap hits and records zero peak wheel momentum in every row; neither establishes actuator adequacy. Initial attitude distribution, magnetic field, gains, source revision, random seeds, timeout and failure reasons are missing. No angular-rate or control-effort histories are supplied. Close the generating model and censoring record before bounded timestep, field-orientation, torque/momentum and coupled acquisition validation. Retain reaction wheels plus magnetorquers; no CMG configuration is inferred.

F.8 Science Record v3 Queue Diagnostic

The corrected sampled-gate workload and outage analysis are in Appendix A.5.5. Native transfer credits above are not v3 records. No arbitrary minimum acquisition duration, count threshold or link derating is introduced. Unknown real acquisition timing and response prevent interpreting the record-slot sensitivity as qualified science delivery.

G Additional MCR Study Evidence and Open Decisions

This appendix identifies the new data files supplied for the MCR. The plots and tabulated results are outputs of the supplied study files and do not establish final payload calibration, a new launch epoch, or compliance with spacecraft requirements by themselves.

G.1 Science grid and mission-tier studies

The MMSC, MSC and EMSC grid-disposition outputs are based on different cell grids and modeled geographic domains. “Primary Map” is the disposition assigned in each output file, not a declaration of global requirement closure. The table summarizes the file contents directly.

Table 38: Science-map cell dispositions in the supplied studies

Study	Evaluated	Primary Map	Supplementary	Unclassified
4-month MMSC	92	92	0	0
6-month MMSC	92	92	0	0
12-month MSC	2020	1915	105	0
14-month MSC	2020	1931	89	0
extended EMSC	7940	4213	3719	8

The MMSC grid file contains 92 modeled cells and the MSC files contain 2020 cells, while the EMSC file contains 7940 cells. These equal the approved polygon-overlap counts at the respective resolutions. They are geometric domains, not automatically eligible science denominators. The missing tier-classification provenance, independent-observation definition and per-channel response still prevent acceptance of the labels as mission success.

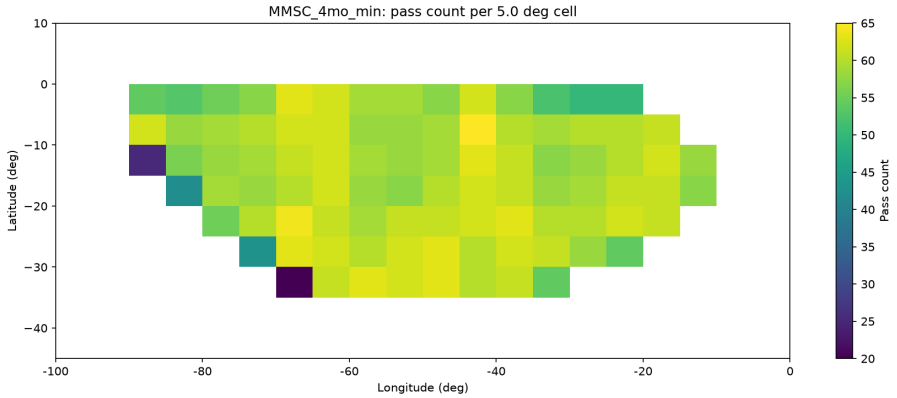


Figure 20: Four-month MMSC science-grid output from the supplied simulation archive; model-dependent.

G.2 Radiation and detector science model

The supplied IRENE-related inputs, proton-spectrum output, stopping-range plots and pile-up calculations support preliminary detector analysis. They do not establish a fully calibrated telescope geometry, dead-time limit, response matrix, or actual AP9 output from the integrated Science simulator. The integrated simulator’s AP9-named quantity is a configured baseline. Current scintillator dimensions are fixed in Appendix A.2; accepted event rates and detector response remain unqualified. The supplied 2026 AP9/AE9 grid is reproduced independently and accepted for preliminary 2029 planning with its original epoch retained. The

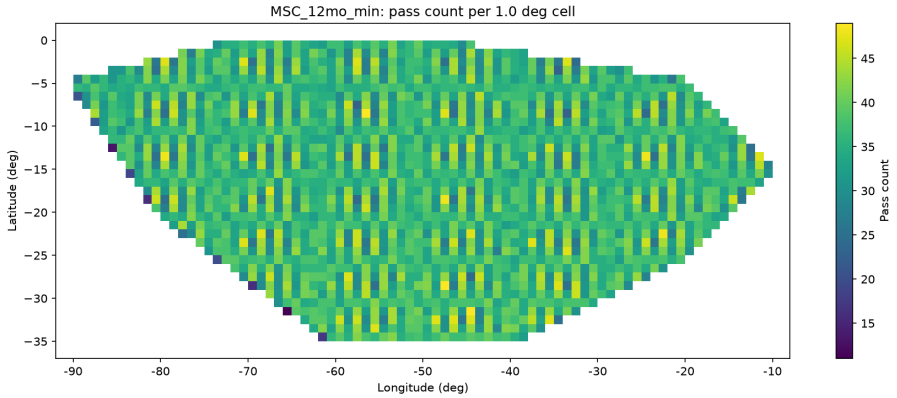


Figure 21: Twelve-month MSC science-grid output from the supplied simulation archive; model-dependent.

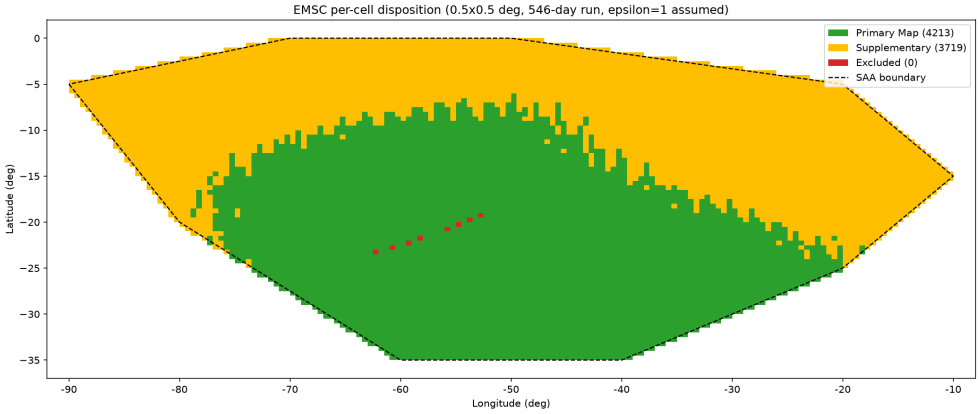


Figure 22: Extended science-grid classification from the supplied simulation archive; model-dependent.

separate November geometric-factor bundle supports the conditional counting requirements in Appendix A.2.5. Its AP9-labelled 65-threshold spectrum has no generating manifest; the verified date of the earlier AP9/AE9 archive must not be transferred to it. The supplied rate heatmap is an ideal integral-above-5 MeV rate proxy, not a detector-efficiency or required-G map.

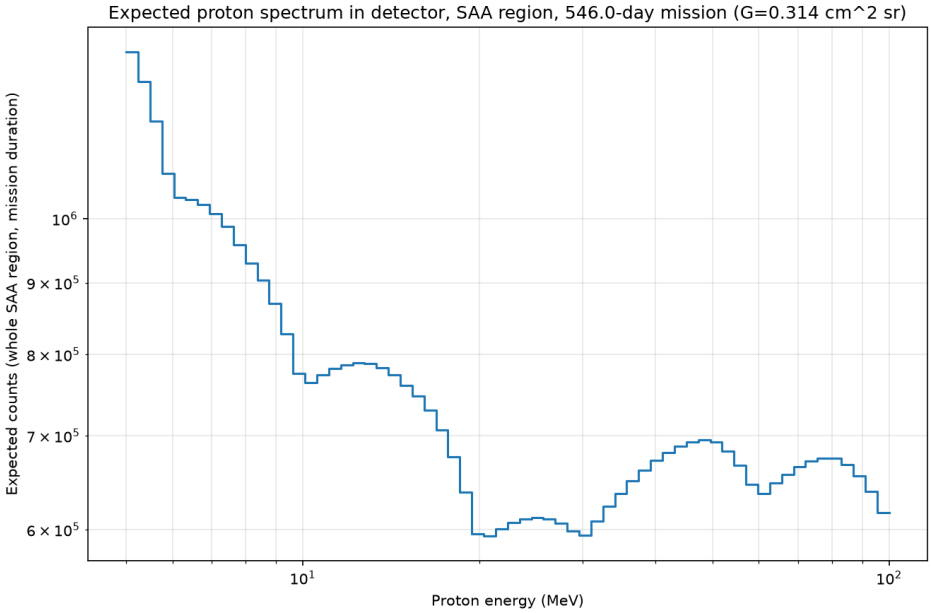


Figure 23: Historical modeled SAA proton spectrum. Its title assumes 546 days and $G = 0.314$ cm^2 sr; these labels do not verify runtime duration or connect it to the new unit-G table. This is not a validated detector measurement.

G.3 Ground-station and DITL source files

The corrected 24-hour and annual exports supersede the earlier incorrect/incomplete Science files. Both day contact logs contain two intervals totaling 766 s. All event origins identify March 2000. Complete recorded horizons do not establish 2029 calibration or current-FPDU qualification. Sections 4.1 and F contain the regenerated accounting and limitations.

G.4 Preliminary procurement and mass allocation

The uploaded bill of materials identifies candidate ADCS, OBC, solar-array, EPS, structure, communications, navigation and antenna parts. The sum of currently specified component-mass entries is 1536.3 g. The payload and deployer-interface hardware do not have mass entries, and the deployable-panel quantity and overall packaging need review. Consequently the bill of materials cannot yet close the mission target of less than 3000 g. Procurement prices remain to be supplied.

G.5 Manufacturer documentation and CAD inputs

Manufacturer documents were supplied for the ISISPACE ICEPS2, TensorTech integrated ADCS, GomSpace NanoMind A3200 and NanoCom AX100, 2nd Space CORE/DROP/VERSE hardware and the proposed GNSS components. CAD inputs were supplied for several candidate components. Final mechanical envelopes, actuator configurations, harness allowances and interface drawings must be checked against the actual proposed configuration before

the component baselines become final.

G.6 Remaining verification conditions

Table 39: Open evidence required for MCR closure

ID	Open item	Needed evidence
O-01	Reconcile historical SAA counter	New two-run orbit classifications are internally consistent; establish classifier geometry and event definitions
O-02	2029 campaign identity and RAAN robustness	Runtime manifests and environment timestamps for the team-accepted reference
O-03	Long-term drag and disposal	Verify new 8–10-year cutoff runs, one truncated case, and post-cutoff reentry
O-04	Solar area, efficiency, degradation and EOL margin	Approved solar/EPS report from final configuration
O-05	FPDU response and electronics	Current CAD, optical isolation, transport, timing and mixed-species trigger validation
O-06	Science-tier coverage requirements	Consistent geographic denominator, per-channel uncertainties and pass requirements
O-07	Ground station RF and data-rate closure	Final payload packets, coding, link and antenna design
O-08	Thermal and structural margins	Model outputs checked against component limits; tests planned
O-09	Complete spacecraft mass and cost	Payload, wiring, deployment items, quantities and quotations
O-10	Integrated long-term and scenario verification	Approved output files, code revision and run parameters

G.7 Bill of materials, as supplied

The following values are transcribed from the uploaded `BILL OF MATERIALS.xlsx`. Components are candidates, and none of the unresolved part numbers or interfaces is to be regarded as a final PDR selection. Values of “TBD” have not been imputed.

Table 40: Candidate component mass entries from supplied bill of materials

Subsystem	Component or part	Qty.	Mass (g)
ADCS	ADCS-10m	1	495
OBC	NanoMind A3200	1	24
Power Generation	Core 1Z	1	114
Power Generation	DROP	1	118
EPS	ICEPS 2S1P+iEPS Battery Type A	1	189
Structure	Verse-03	1	310
Communications	NanoCom AX100	1	24.5
Communications	UHF dipole antenna	1	90
Communications	ANT430	1	30
Deployment	TBD	1	TBD
Payload	Custom	1	TBD
Navigation	ZED-F9P-15B	1	1.8

Subsystem	Component or part	Qty.	Mass (g)
Navigation	AHP54510.07.0100C	1	140

The workbook reports a total of 1536.3 g of currently entered masses, before adding the unallocated payload and deployment-interface mass. The current solar-panel quantity entries require confirmation against the intended one fixed and two deployable panel geometry. Costs in the workbook are not quoted amounts.

G.8 Datasheet alignment checks

The supplied ICEPS2 datasheet describes a 2S1P battery pack with nominal energy 28.8 Wh. The 25 Wh value remains the mission simulation planning capacity. The supplied TensorTech ADCS-10m datasheet explicitly describes a control-moment gyroscope configuration. Since ANTARA intends to use a reaction-wheel and magnetorquer integrated system, that datasheet cannot be treated as proof of the selected ADCS hardware configuration. A matching vendor model and specification must be supplied. The GNSS, radio, antenna, deployment-panel and structure datasheets are candidate mechanical/electrical references, pending interface selection.

G.9 Approved 2029 Planning Baseline

The mission reference is 1 January 2029: 3U, maximum 3.0 kg, approximately 600 km SSO, 97.6° inclination, eccentricity approximately 0.001, launch-dependent RAAN, 25 Wh battery planning capacity and reaction wheels with magnetorquers. The team accepts the supplied RAAN data for 2029 planning; the inspected local MCR runtime still starts in 2026, so its generating epoch remains unverified. This does not prevent use as a clearly identified preliminary reference. The supplied radiation map retains its verified 2026 source epoch.

The approved FPDU has five longitudinal main blocks and one front scintillator, eleven SiPMs, no silicon entrance and optional rear detection. Optical isolation, material, calibrated response and processor selection remain open. Appendix A.2 controls current construction and timing. Approximately five years is the operational objective; modeled decay to 300 km and subsequent reentry are separate quantities. The 8–10-year cutoff study retains its configuration and censoring limitations (Appendix H).

H Updated RAAN Trade and Orbital Decay Evidence

This appendix transcribes the four latest supplied MCR campaign CSV files: `raan_master_trade_matrix(1).csv`, `mission_statistics(1).csv`, `drag_lifetime_analysis(1).csv`, and `orbit_classification(1).csv`. These outputs are the newest *provisional* trade-study evidence, not an integrated DITL rerun or a flight-verified performance guarantee. The approved mission analysis epoch is 1 January 2029; the supplied CSV files alone do not prove that they were generated with that epoch or with the final 3 kg spacecraft model.

H.1 RAAN trade-study summary

Table 41: Latest RAAN trade matrix, as provided. SOC is expressed as a percentage of capacity; the minimum column averages run minima, not the worst sample; orbit counts are sums across the study runs, not annual values.

RAAN	Power (W)	Mean SOC (%)	Mean min. (%)	Life (yr)	SAA-only	GS-only
0°	11.02	97.9	69.6	9.46	79,377	11,536
30°	10.82	97.2	74.7	9.53	79,865	11,613
60°	10.28	95.1	70.4	8.95	74,917	10,995
90°	10.13	94.9	72.7	8.88	74,340	10,969
120°	9.02	88.1	54.4	8.23	69,267	10,196
150°	9.50	92.7	71.6	8.66	72,819	10,657
180°	11.17	98.2	73.4	8.58	72,103	10,498

The sampled RAAN set is $0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ, 180^\circ$. The largest tabulated mean generated power is 11.17 W at 180°; the smallest is 9.02 W at 120°. The latter has the lowest mean of run-minimum SOC (54.4%). Consequently the current study does *not* justify naming 0° or 30° universally optimal. The choice of RAAN is launch-dependent; a robust spacecraft must be assessed across its feasible orbital-insertion cases. The RAAN matrix does not by itself demonstrate the fully integrated power/attitude/communications performance of a final 2029 mission.

H.2 Orbit-classification bookkeeping

The raw classification table contains 685798 rows: seven RAAN values and two recorded runs per RAAN. The entries in the RAAN trade matrix match the summed counts of `QUIET`, `ONLY_GS`, `ONLY_SAA` and `MIXED` orbit classes in that source file. `orbit_number` is unique within each RAAN/run pair. Classifications are mutually exclusive for each orbit record, but the tabulated counts pool both runs and therefore must not be presented as a one-year orbit count. The modeled classifier and event geometry still require documentation before they can be used as physical detector-on SAA passes or real GS contacts.

H.3 Lifetime to the modeled 300 km cutoff

Across the 14 records, modeled termination times span 8.23–9.99 years. Thirteen rows terminate near 300 km. The remaining 30° record terminates at 334.93 km after approximately 9.99 years, so it cannot be counted as a completed decay-to-300-km result. This suggests an approximately 8–10-year decay timescale *under these study assumptions*, not proof of the earlier five-year orbital-decay estimate. The output includes no post-300-km propagation to an atmospheric reentry condition. Atmospheric model, ballistic coefficient, starting epoch,

Table 42: Two drag-study records per RAAN. Time is to termination of the supplied integration; the listed endpoint is not atmospheric reentry.

RAAN	Recorded time (yr), range	Endpoint altitude (km), range
0°	8.96–9.96	299.9–299.9
30°	9.06–9.99	299.9–334.9
60°	8.87–9.04	300.0–300.0
90°	8.86–8.90	299.6–299.8
120°	8.23–8.24	300.0–300.0
150°	8.53–8.79	299.9–299.9
180°	8.33–8.83	299.9–299.9

mass, attitude-dependent area, solar activity, termination cause and parameter uncertainty require source-code provenance review. These results are preliminary and do not establish disposal compliance.

H.4 Monte Carlo statistics: scope and interpretation

The provided `mission_statistics(1).csv` corresponds numerically to the 180° RAAN row for average power (11.17485 W), mean SOC (0.98234), mean run-minimum SOC (0.73446), cycle count (6565.43) and lifetime (8.58241 years). Its confidence intervals should not be interpreted as 99% population coverage without verification of the statistical method and independent sample count; the raw study includes only two runs per RAAN. The power and lifetime fields represent modeled quantities, not measured solar electrical output or a verified physical lifetime.

H.5 Design implication for the MCR

The operational science campaign targets a roughly five-year mission, while the newest drag study points toward a potentially longer residence time before reaching 300 km. These statements are not interchangeable. The MCR should maintain the five-year *operational objective* while qualifying the uncertain passive decay and post-mission disposal timeline. Detailed DITL operations are being rerun separately; their future results should be incorporated with independent identification of date, simulation revision and model assumptions.